



**Politecnico
di Torino**



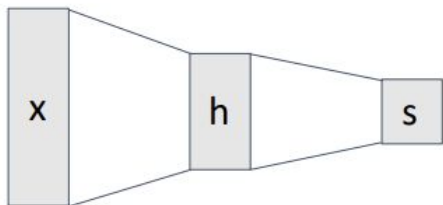
Advanced Machine Learning

A.A. 2023/2024

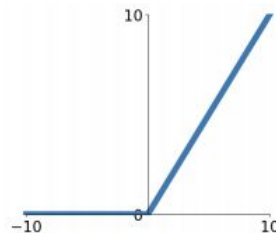
Tatiana Tommasi

Components of a Convolutional Network

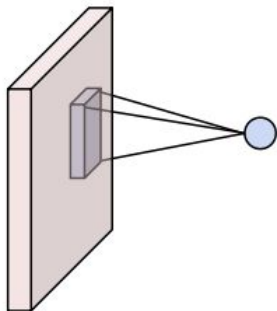
Fully-Connected Layers



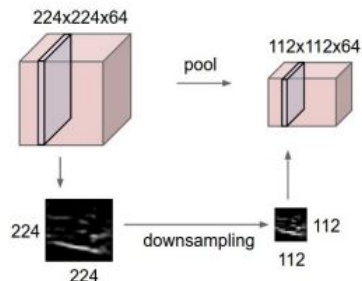
Activation Function



Convolution Layers



Pooling Layers

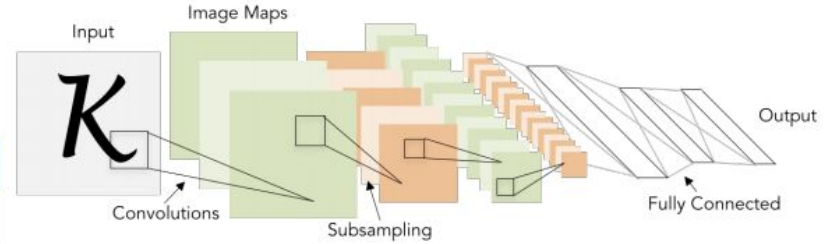


Normalization

$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$$

Example: LeNet-5

Layer	Output Size	Weight Size
Input	1 x 28 x 28	
Conv ($C_{out}=20$, $K=5$, $P=2$, $S=1$)	20 x 28 x 28	20 x 1 x 5 x 5
ReLU	20 x 28 x 28	
MaxPool($K=2$, $S=2$)	20 x 14 x 14	
Conv ($C_{out}=50$, $K=5$, $P=2$, $S=1$)	50 x 14 x 14	50 x 20 x 5 x 5
ReLU	50 x 14 x 14	
MaxPool($K=2$, $S=2$)	50 x 7 x 7	
Flatten	2450	
Linear (2450 -> 500)	500	2450 x 500
ReLU	500	
Linear (500 -> 10)	10	500 x 10



As we go through the network:

Spatial size **decreases**
(using pooling or strided conv)

Number of channels **increases**
(total “volume” is preserved!)

Lecun et al, “Gradient-based learning applied to document recognition”, 1998

Batch Normalization

Idea: “Normalize” the outputs of a layer so they have zero mean and unit variance

Why? Helps reduce “internal covariate shift”, improves optimization

We can normalize a batch of activations like this:

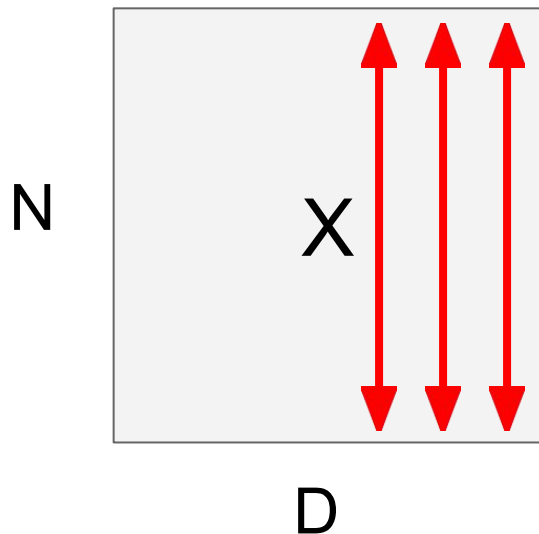
$$\hat{x} = \frac{x - E[x]}{\sqrt{Var[x]}}$$

This is a **differentiable function**, so we can use it as an operator in our networks and backprop through it!

Ioffe and Szegedy, “Batch normalization: Accelerating deep network training by reducing internal covariate shift”, ICML 2015

Batch Normalization

“you want unit gaussian activations?
just make them so.”



Compute the empirical mean and variance independently for each dimension.

$$\mu_j = \frac{1}{N} \sum_{i=1}^N x_{i,j} \quad \text{Per-channel mean, shape is D}$$

$$\sigma_j^2 = \frac{1}{N} \sum_{i=1}^N (x_{i,j} - \mu_j)^2 \quad \text{Per-channel std, shape is D}$$

$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}} \quad \text{Normalized x, Shape is N x D}$$

Batch Normalization

Normalize:

$$\hat{x}^{(k)} = \frac{x^{(k)} - \mathbb{E}[x^{(k)}]}{\sqrt{\text{Var}[x^{(k)}]}}$$

And then allow the network to squash the range if it wants to:

$$y^{(k)} = \gamma^{(k)} \hat{x}^{(k)} + \beta^{(k)}$$

Note, the network can learn:

$$\gamma^{(k)} = \sqrt{\text{Var}[x^{(k)}]}$$

$$\beta^{(k)} = \mathbb{E}[x^{(k)}]$$

to recover the identity mapping.

Batch Normalization

[Ioffe and Szegedy, 2015]

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1\dots m}\}$;
Parameters to be learned: γ, β

Output: $\{y_i = \text{BN}_{\gamma, \beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad // \text{ mini-batch mean}$$

$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_{\mathcal{B}})^2 \quad // \text{ mini-batch variance}$$

$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \quad // \text{ normalize}$$

$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \quad // \text{ scale and shift}$$

- Improves gradient flow through the network
- Allows higher learning rates
- Reduces the strong dependence on initialization
- Acts as a form of regularization

Batch Normalization

[Ioffe and Szegedy, 2015]

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1\dots m}\}$;
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Note: at test time BatchNorm layer functions differently:

The mean/std are not computed based on the batch. Instead, a single fixed empirical mean of activations during training is used.

(e.g. can be estimated during training with running averages)

Batch Normalization

[Ioffe and Szegedy, 2015]

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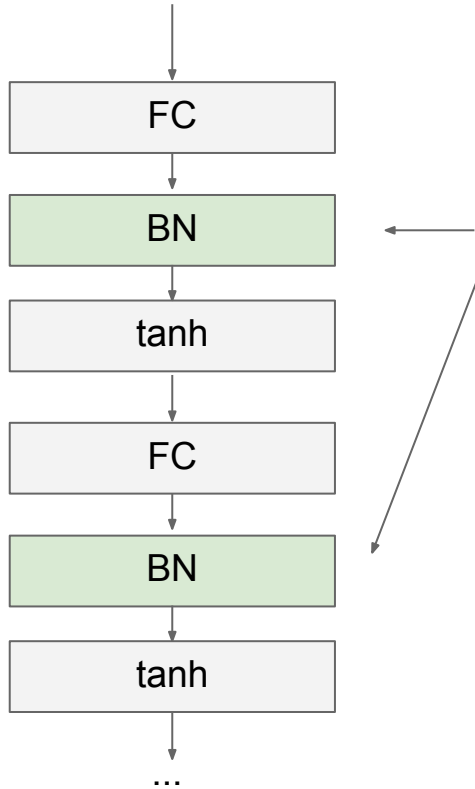
Note: at test time BatchNorm layer functions differently:

The mean/std are not computed based on the batch. Instead, a single fixed empirical mean of activations during training is used.

(e.g. can be estimated during training with running averages)

During testing batchnorm becomes a linear operator!
Can be fused with the previous fully-connected or conv layer

Batch Normalization



Usually inserted after Fully Connected or Convolutional layers, and before nonlinearity.

$$\hat{x}^{(k)} = \frac{x^{(k)} - \mathbb{E}[x^{(k)}]}{\sqrt{\text{Var}[x^{(k)}]}}$$

Batch Normalization

Batch Normalization for
fully-connected networks

$$x : N \times D$$

Normalize

$$\mu, \sigma : 1 \times D$$

$$\gamma, \beta : 1 \times D$$

$$y = \frac{(x - \mu)}{\sigma} \gamma + \beta$$

Batch Normalization for
convolutional networks
(Spatial Batchnorm, BatchNorm2D)

$$x : N \times C \times H \times W$$

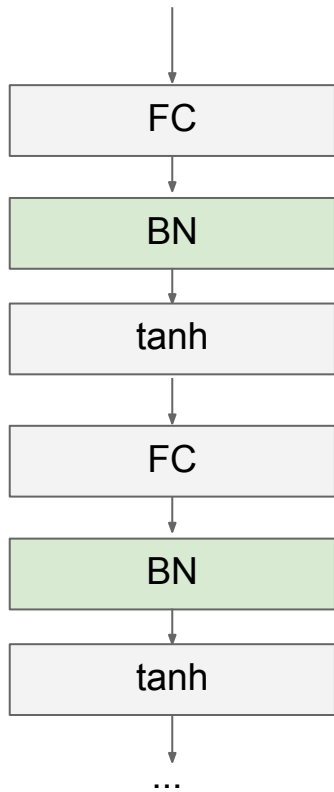
Normalize

$$\mu, \sigma : 1 \times C \times 1 \times 1$$

$$\gamma, \beta : 1 \times C \times 1 \times 1$$

$$y = \frac{(x - \mu)}{\sigma} \gamma + \beta$$

Batch Normalization



- Makes deep networks **much** easier to train!
- Allows higher learning rates, faster convergence
- Networks become more robust to initialization
- Acts as regularization during training
- Zero overhead at test-time: can be fused with conv!
- **Not well-understood theoretically (yet)**
- **Behaves differently during training and testing: this is a very common source of bugs!**

Layer Normalization

Batch Normalization for
fully-connected networks

$$x : N \times D$$

Normalize

$$\mu, \sigma : 1 \times D$$

$$\gamma, \beta : 1 \times D$$

$$y = \frac{(x - \mu)}{\sigma} \gamma + \beta$$

Layer Normalization for fully-
connected networks
Same behavior at train and test!
Used in RNNs, Transformers

$$x : N \times D$$

Normalize

$$\mu, \sigma : N \times 1$$

$$\gamma, \beta : 1 \times D$$

$$y = \frac{(x - \mu)}{\sigma} \gamma + \beta$$

Instance Normalization

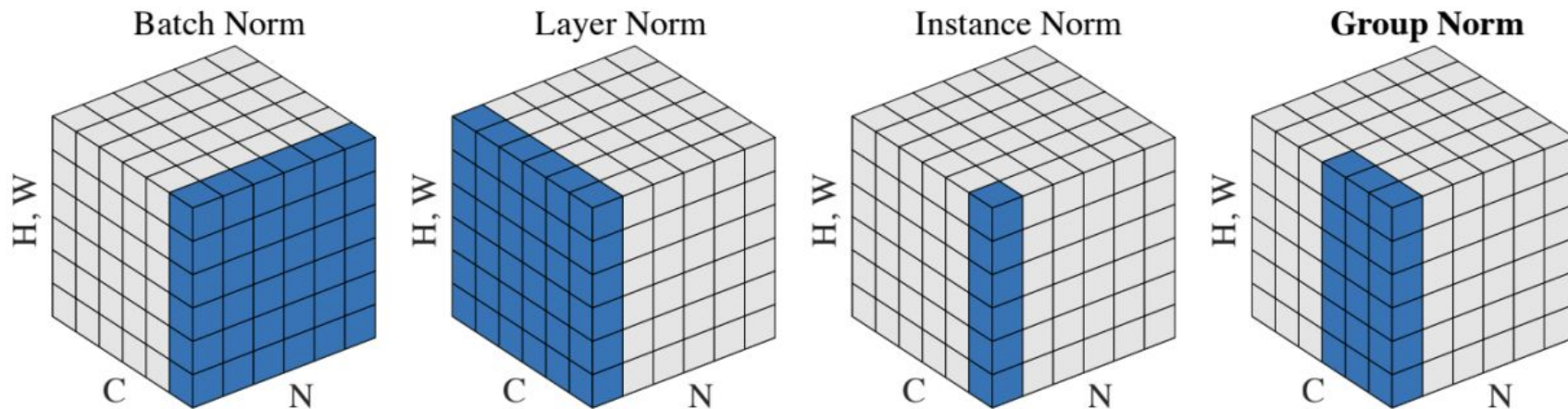
Batch Normalization for
convolutional networks

$$\begin{array}{l} x : N \times C \times H \times W \\ \text{Normalize} \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow \\ \mu, \sigma : 1 \times C \times 1 \times 1 \\ \gamma, \beta : 1 \times C \times 1 \times 1 \\ y = \frac{(x - \mu)}{\sigma} \gamma + \beta \end{array}$$

Instance Normalization for
convolutional networks

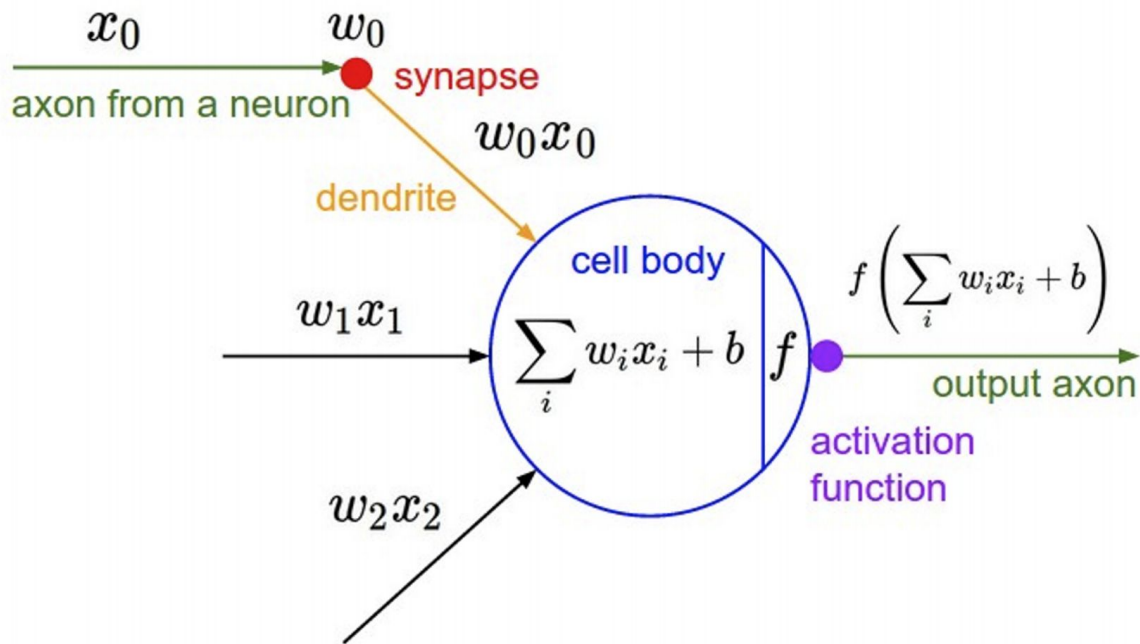
$$\begin{array}{l} x : N \times C \times H \times W \\ \text{Normalize} \quad \quad \quad \downarrow \quad \quad \downarrow \\ \mu, \sigma : N \times C \times 1 \times 1 \\ \gamma, \beta : 1 \times C \times 1 \times 1 \\ y = \frac{(x - \mu)}{\sigma} \gamma + \beta \end{array}$$

Comparison of Normalization Layers



Wu and He, "Group Normalization", ECCV 2018

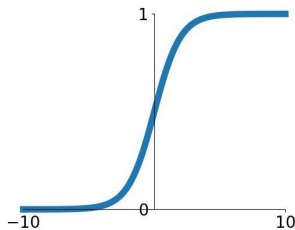
Activation Functions



Activation Functions

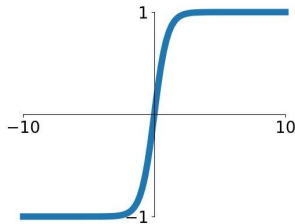
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



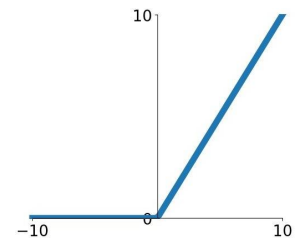
tanh

$$\tanh(x)$$



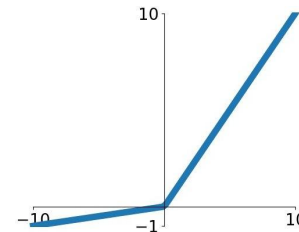
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

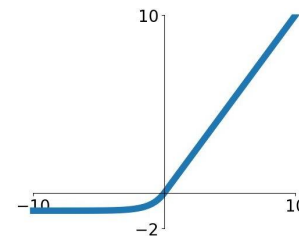


Maxout

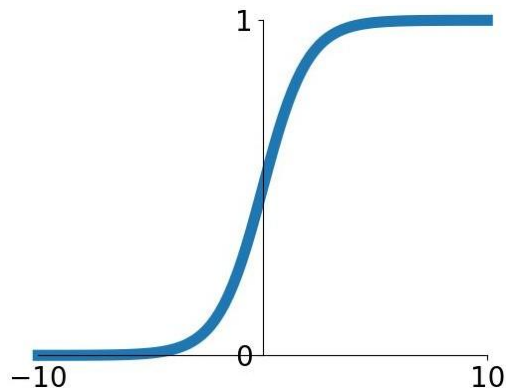
$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



Activation Functions

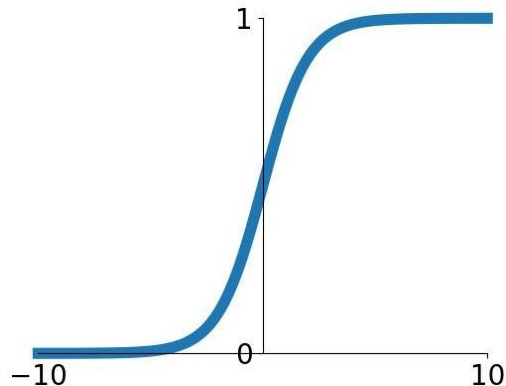


Sigmoid

$$\sigma(x) = 1/(1 + e^{-x})$$

- Squashes numbers to range $[0, 1]$
- Historically popular since they have nice interpretation as a saturating “firing rate” of a neuron

Activation Functions



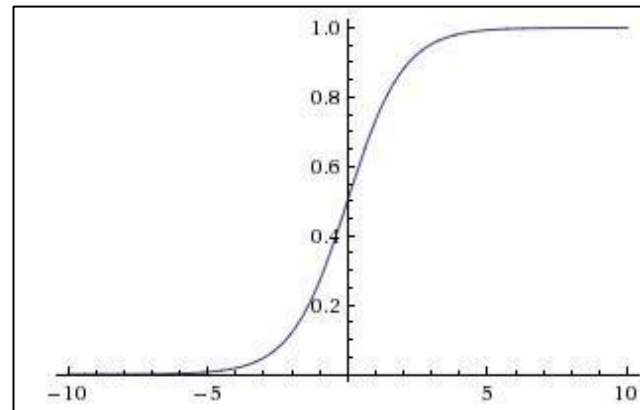
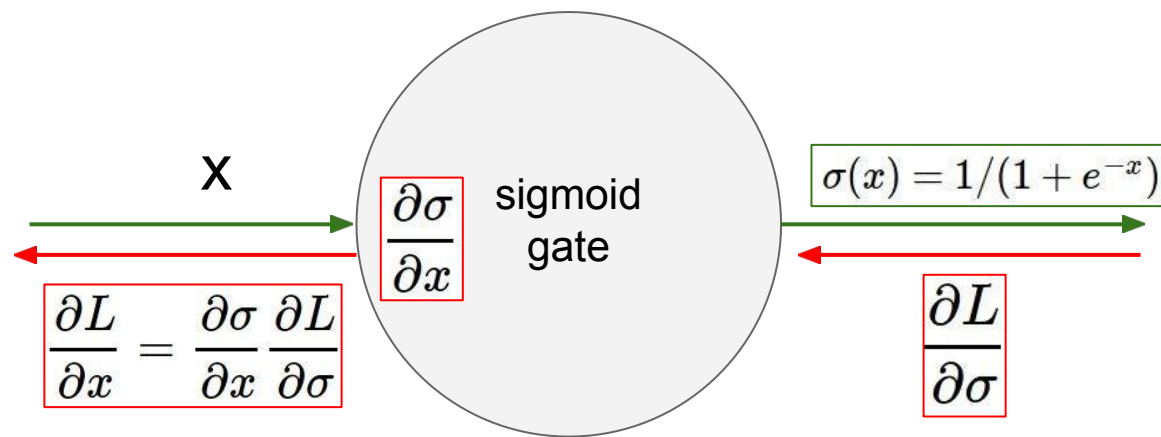
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3 problems:

1. Saturated neurons “kill” the gradients

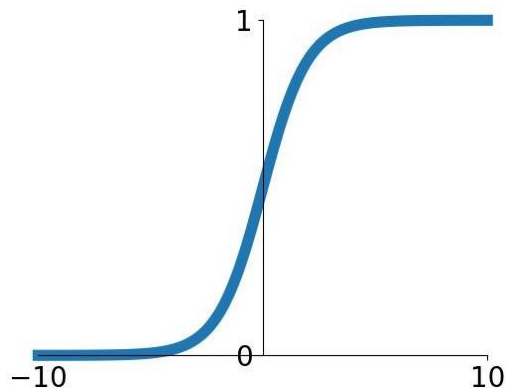


What happens when $x = -10$?

What happens when $x = 0$?

What happens when $x = 10$?

Activation Functions



Sigmoid

$$\sigma(x) = 1/(1 + e^{-x})$$

- Squashes numbers to range $[0,1]$
- Historically popular since they have nice interpretation as a saturating “firing rate” of a neuron

3 problems:

1. Saturated neurons “kill” the gradients
2. Sigmoid outputs are not zero-centered

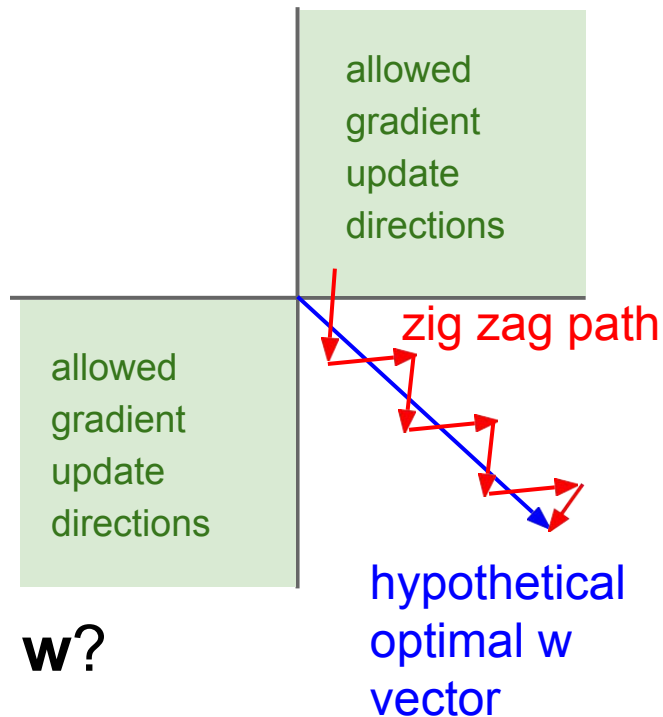
Consider what happens when the input to a neuron (x) is always positive:

$$f = \sum w_i x_i + b$$

$$\frac{df}{dw_i} = x_i$$

$$\frac{dL}{dw_i} = \frac{dL}{df} \frac{df}{dw_i} = \frac{dL}{df} x_i$$

because $x_i > 0$, the gradient $\frac{dL}{dw_i}$ always has the same sign as $\frac{dL}{df}$ (all positive or all negative).

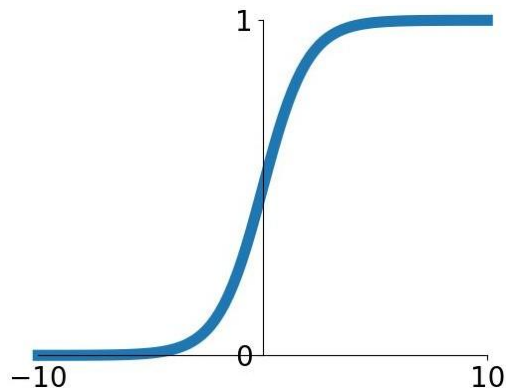


What can we say about the gradients on $\mathbf{w}$?

Always all positive or all negative :(

(issue for 1 element. Mini-batches help)

Activation Functions



Sigmoid

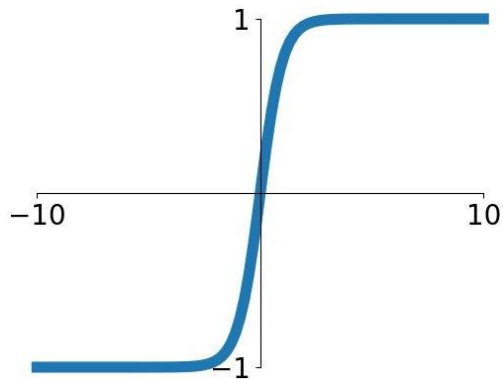
$$\sigma(x) = 1/(1 + e^{-x})$$

- Squashes numbers to range $[0, 1]$
- Historically popular since they have nice interpretation as a saturating “firing rate” of a neuron

3 problems:

1. Saturated neurons “kill” the gradients
2. Sigmoid outputs are not zero-centered
3. $\exp()$ is a bit compute expensive

Activation Functions

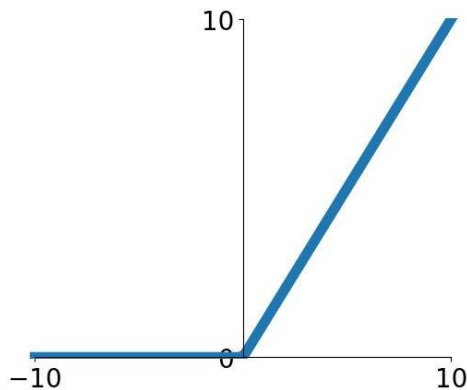


$\tanh(x)$

- Squashes numbers to range $[-1,1]$
- zero centered (nice)
- still kills gradients when saturated :(

[LeCun et al., 1991]

Activation Functions



Computes $f(x) = \max(0, x)$

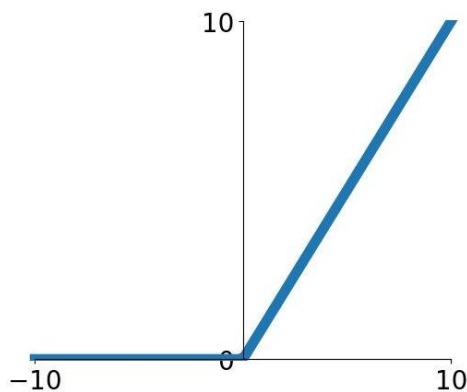
- Does not saturate (in +region)
- Very computationally efficient
- Converges much faster than sigmoid/tanh in practice (e.g. 6x)
- Actually more biologically plausible than sigmoid

ReLU

(Rectified Linear Unit)

[Krizhevsky et al., 2012]

Activation Functions



Computes $f(x) = \max(0, x)$

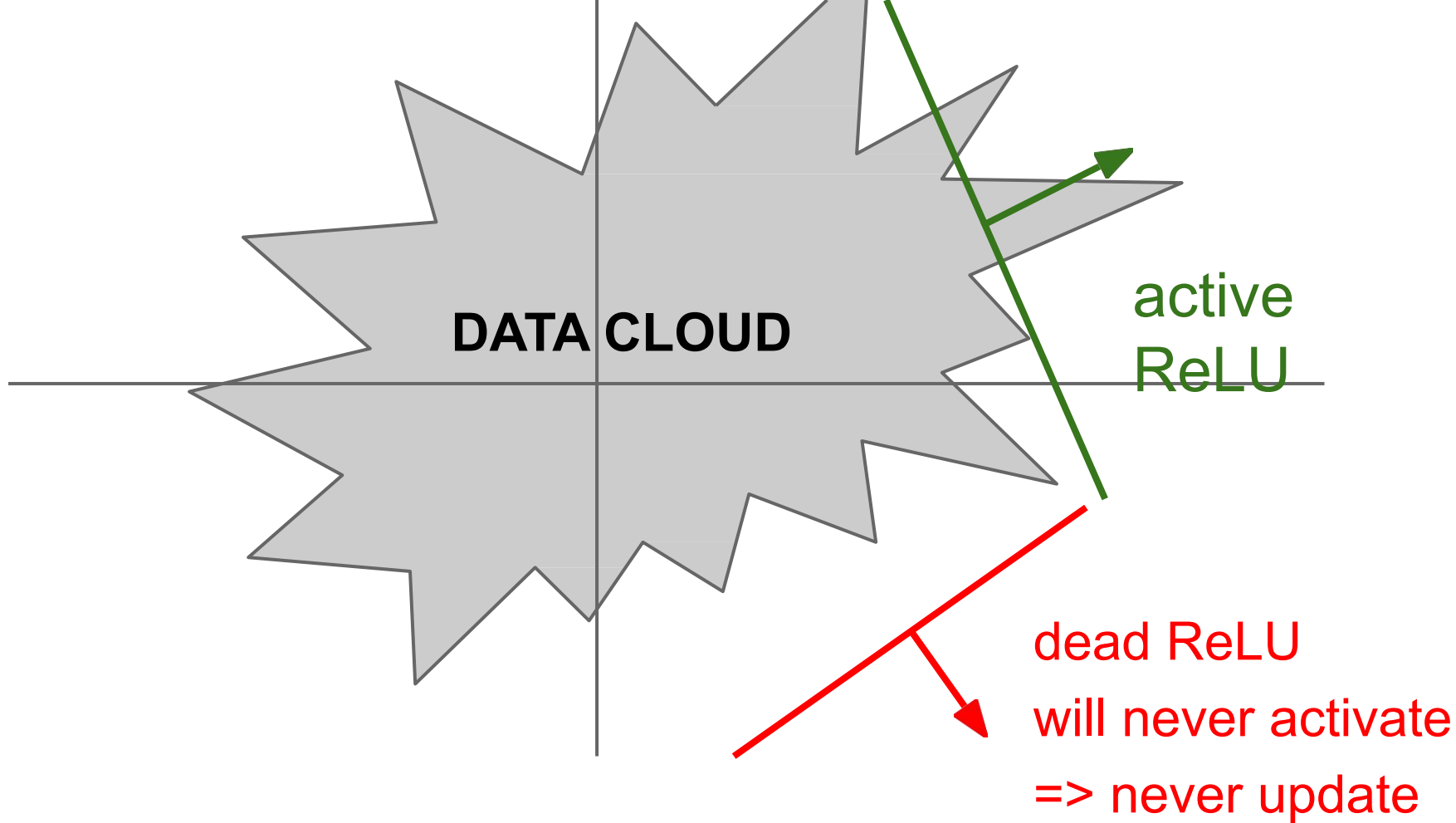
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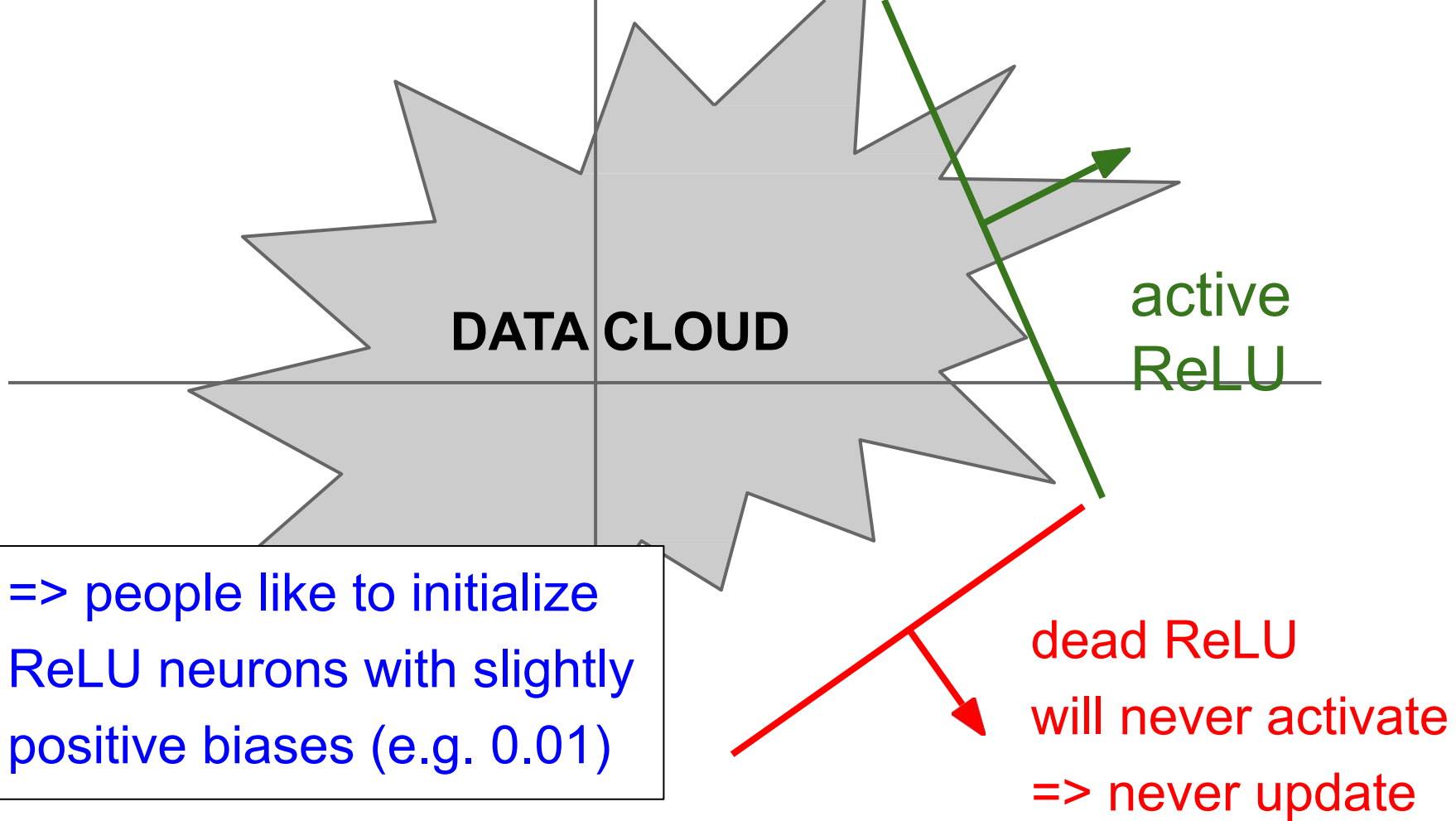
ReLU

(Rectified Linear Unit)

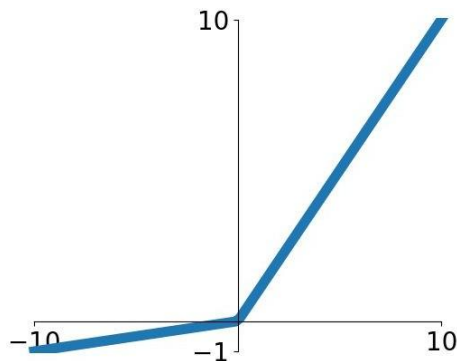
Not zero-centered output

An annoyance: what is the gradient when $x < 0$?





Activation Functions

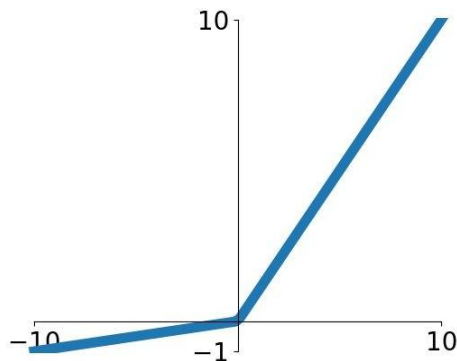


- Does not saturate
- Computationally efficient
- Converges much faster than sigmoid/tanh in practice! (e.g. 6x)
- **will not “die”.**

Leaky ReLU

$$f(x) = \max(0.01x, x)$$

Activation Functions



Leaky ReLU

$$f(x) = \max(0.01x, x)$$

- Does not saturate
- Computationally efficient
- Converges much faster than sigmoid/tanh in practice! (e.g. 6x)
- **will not “die”.**

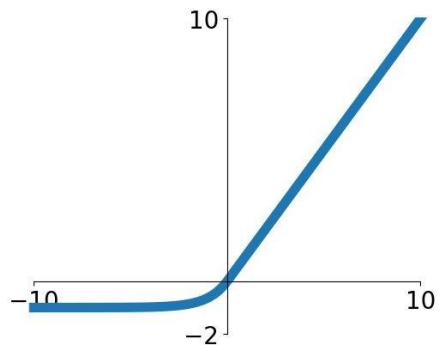
Parametric Rectifier (PReLU)

$$f(x) = \max(\alpha x, x)$$

backprop into α (parameter)

Activation Functions

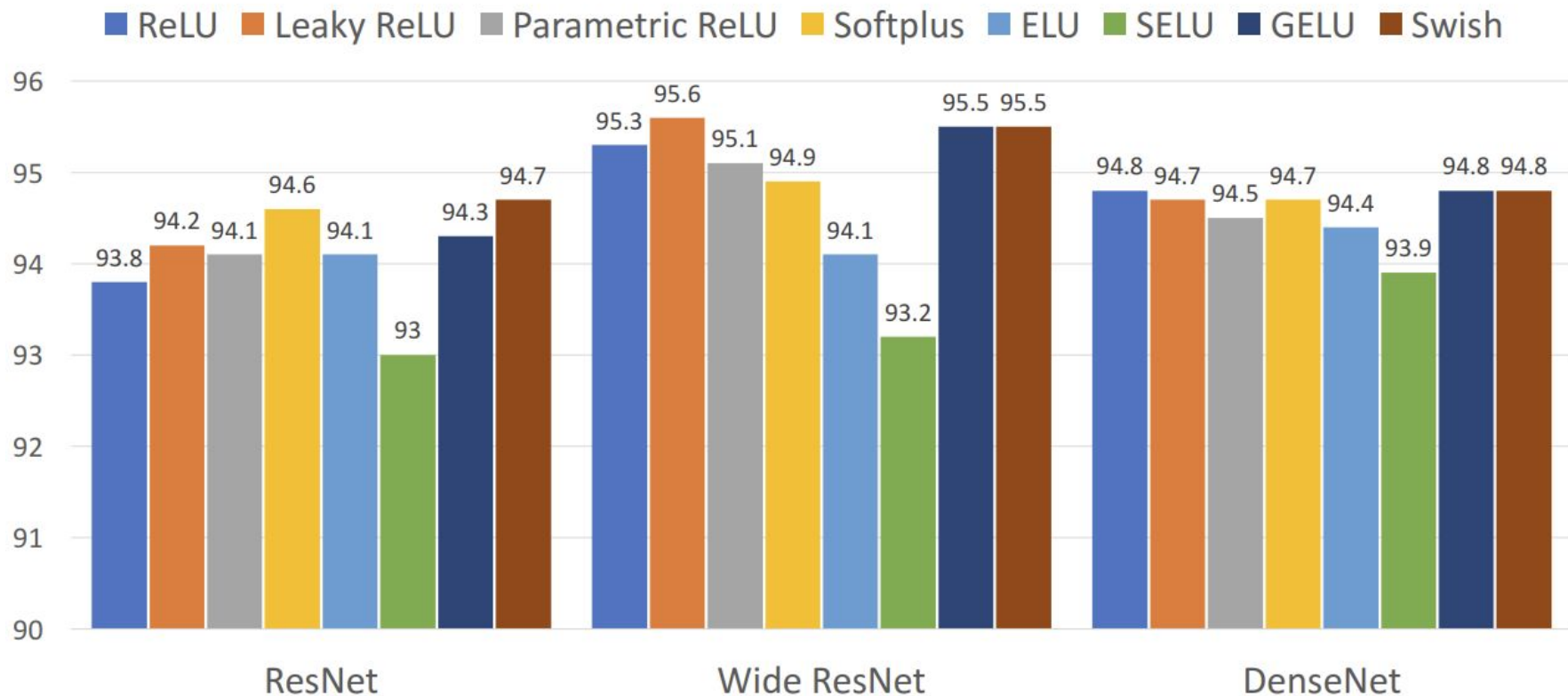
Exponential Linear Units (ELU)



$$f(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha (\exp(x) - 1) & \text{if } x \leq 0 \end{cases}$$

- All benefits of ReLU
- Closer to zero mean outputs
- Negative saturation regime compared with Leaky ReLU adds some robustness to noise
- Computation requires $\exp()$

Accuracy on CIFAR10



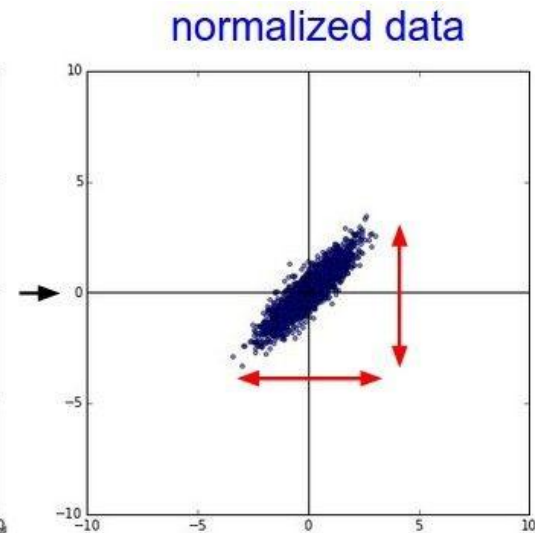
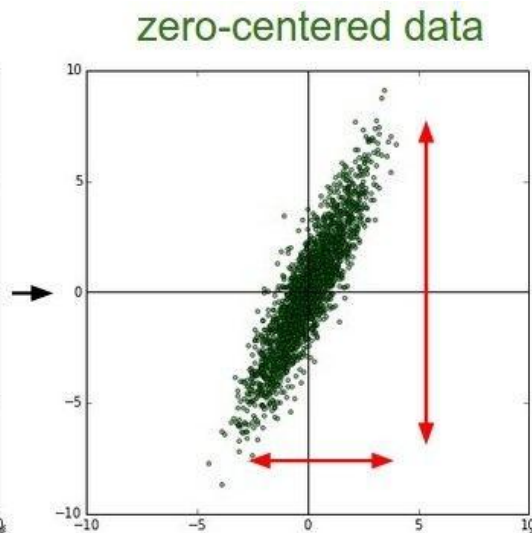
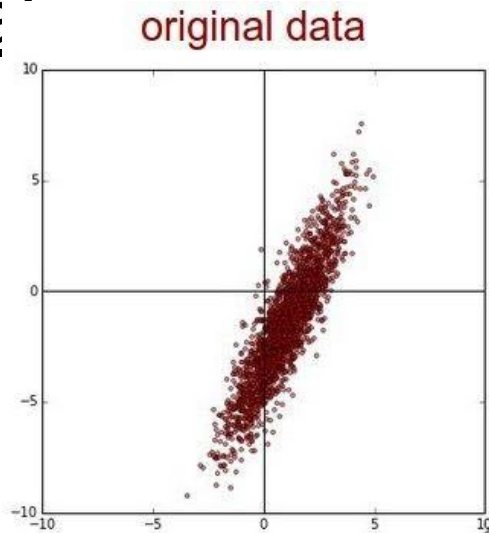
Rule of Thumb: in practice

- Don't think too hard. Just use **ReLU**.
- Try out **Leaky ReLU / ELU (SELU)**
if you need to squeeze that last 0.1%
- **Don't use sigmoid or tanh**

Data Preprocessing

Step 1: Preprocess the

data



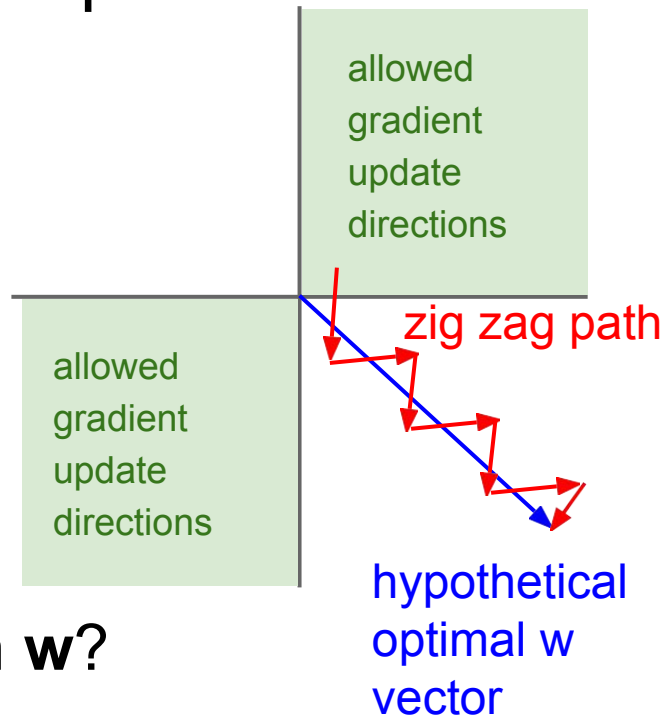
```
X -= np.mean(X, axis = 0)
```

```
X /= np.std(X, axis = 0)
```

(Assume X [NxD] is data matrix,
each example in a row)

Consider what happens when the input to a neuron (x) is always positive:

$$f\left(\sum_i w_i x_i + b\right)$$



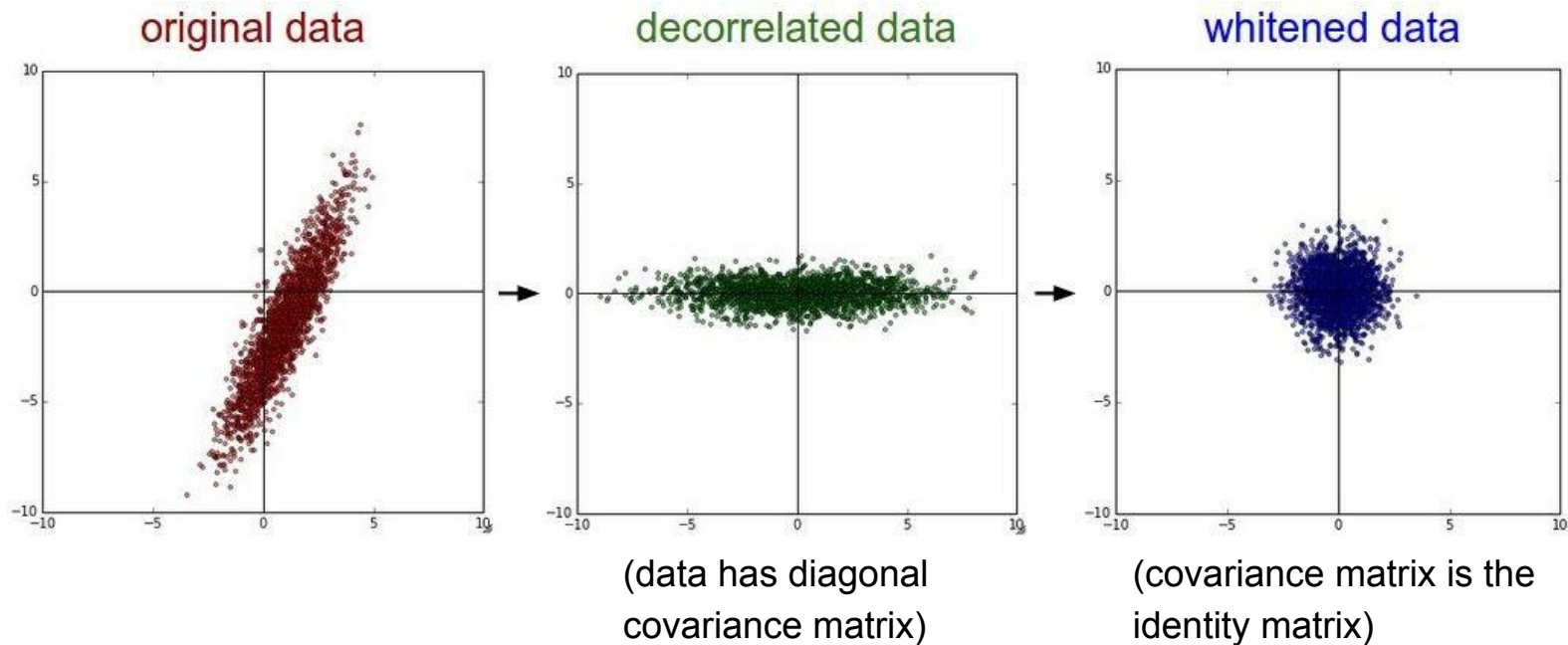
What can we say about the gradients on $\mathbf{w}$?

Always all positive or all negative :(

(this is also why you want zero-mean data!)

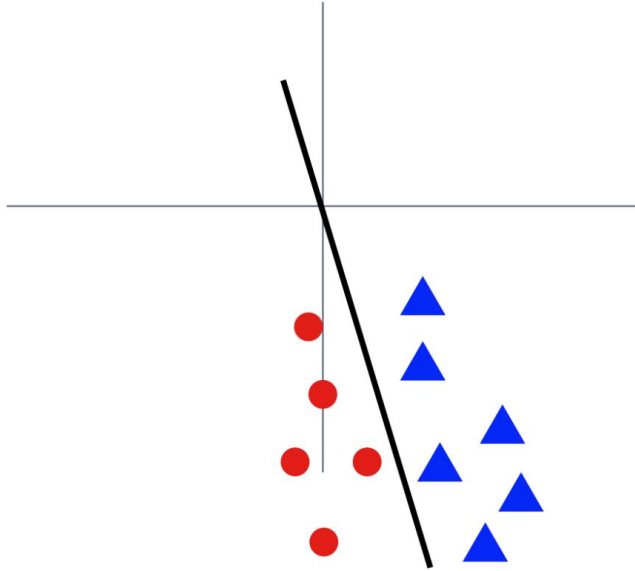
Step 1: Preprocess the data

In practice, you may also see **PCA** and **Whitening** of the data

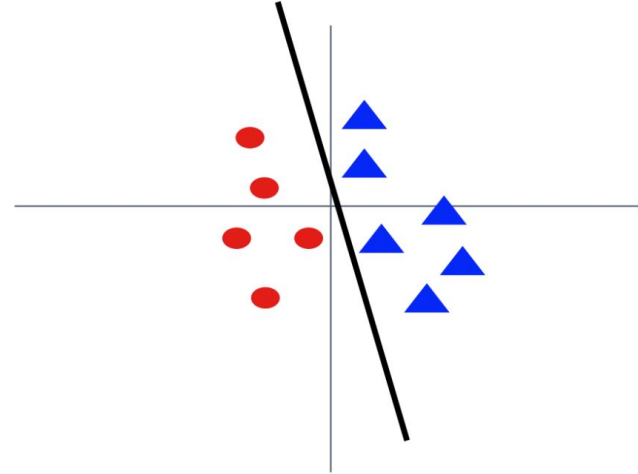


Step 1: Preprocess the data

Before normalization: classification loss very sensitive to changes in weight matrix; hard to optimize



After normalization: less sensitive to small changes in weights; easier to optimize



Rule of Thumb for images = center only

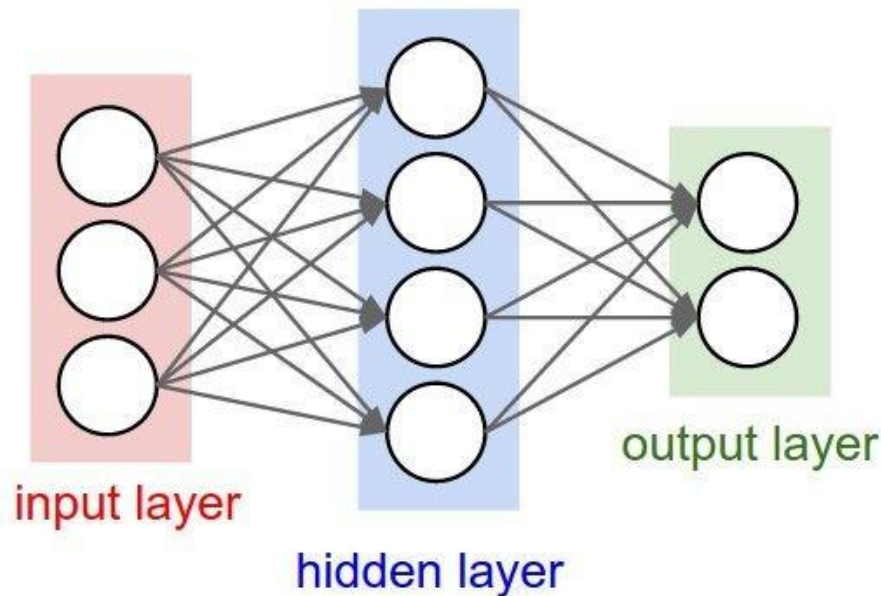
e.g. consider CIFAR-10 example with [32,32,3] images

- Subtract the mean image (e.g. AlexNet)
(mean image = [32,32,3] array)
- Subtract per-channel mean (e.g. VGGNet)
(mean along each channel = 3 numbers)
- Subtract per-channel mean and Divide by per-channel std (e.g. ResNet)
(mean along each channel = 3 numbers)

Not common
to normalize
variance, to
do PCA or
whitening

Weight Initialization

Q: what happens when $W=0$ init is used
(and $b=0$)?



A: All outputs are 0, all
gradients are the same!
No “symmetry breaking”

- First idea: **Small random numbers**
(gaussian with zero mean and $1e-2$ standard deviation)

```
W = 0.01 * np.random.randn(Din, Dout)
```

- First idea: **Small random numbers**
(gaussian with zero mean and $1e-2$ standard deviation)

```
W = 0.01 * np.random.randn(Din, Dout)
```

Works ~okay for small networks, but problems with deeper networks.

Weight Initialization: Activation Statistics

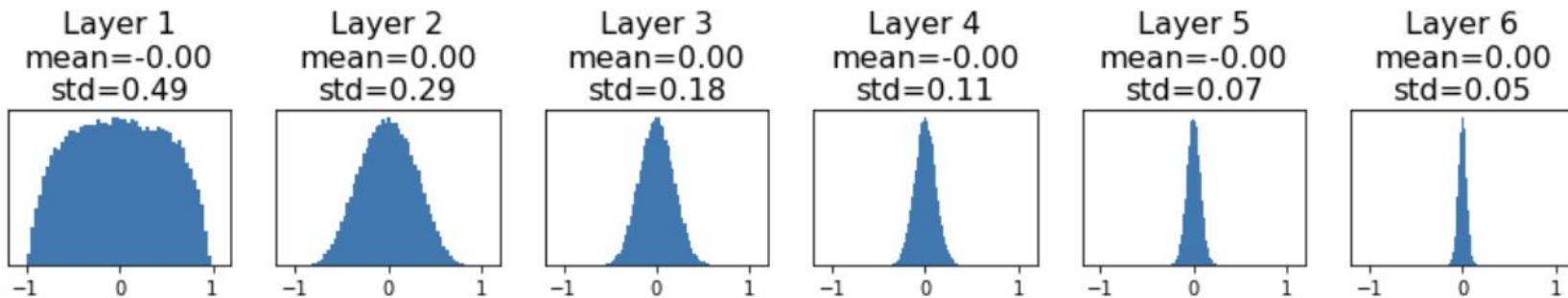
```
dims = [4096] * 7      Forward pass for a 6-layer
hs = []                net with hidden size 4096
x = np.random.randn(16, dims[0])
for Din, Dout in zip(dims[:-1], dims[1:]):
    W = 0.01 * np.random.randn(Din, Dout)
    x = np.tanh(x.dot(W))
    hs.append(x)
```

Weight Initialization: Activation Statistics

```
dims = [4096] * 7      Forward pass for a 6-layer  
hs = []               net with hidden size 4096  
x = np.random.randn(16, dims[0])  
for Din, Dout in zip(dims[:-1], dims[1:]):  
    W = 0.01 * np.random.randn(Din, Dout)  
    x = np.tanh(x.dot(W))  
    hs.append(x)
```

All activations tend to zero for deeper network layers

Q: What do the gradients dL/dW look like?



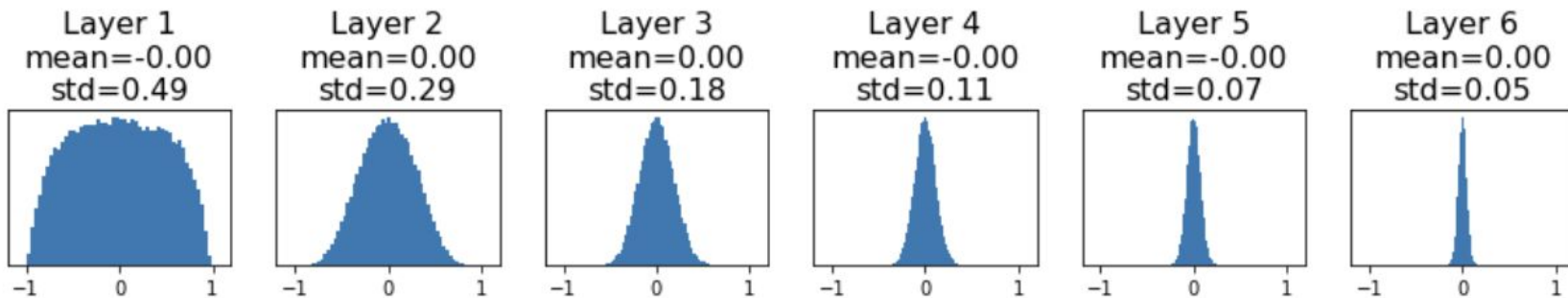
Weight Initialization: Activation Statistics

```
dims = [4096] * 7      Forward pass for a 6-layer
hs = []                net with hidden size 4096
x = np.random.randn(16, dims[0])
for Din, Dout in zip(dims[:-1], dims[1:]):
    W = 0.01 * np.random.randn(Din, Dout)
    x = np.tanh(x.dot(W))
    hs.append(x)
```

All activations tend to zero for deeper network layers

Q: What do the gradients dL/dW look like?

A: All zero, no learning =(



Weight Initialization: Activation Statistics

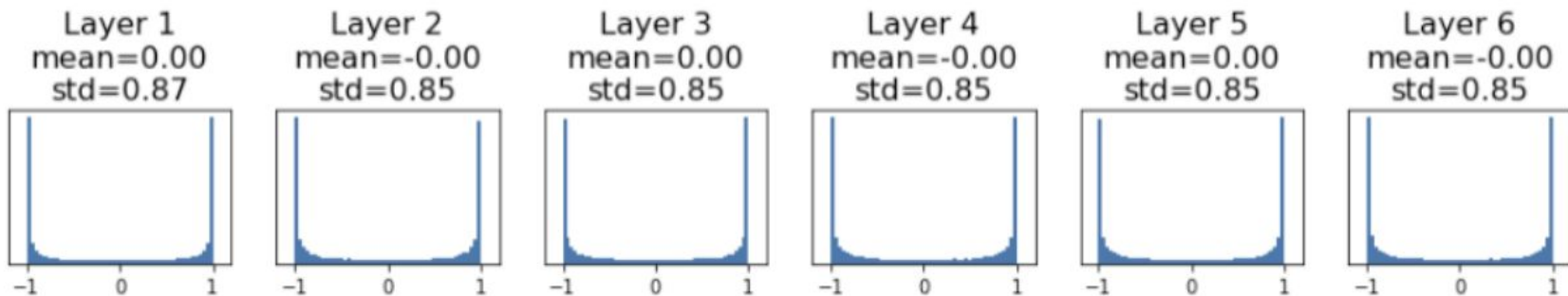
```
dims = [4096] * 7    Increase std of initial weights  
hs = []              from 0.01 to 0.05  
x = np.random.randn(16, dims[0])  
for Din, Dout in zip(dims[:-1], dims[1:]):  
    W = 0.05 * np.random.randn(Din, Dout)  
    x = np.tanh(x.dot(W))  
    hs.append(x)
```

Weight Initialization: Activation Statistics

```
dims = [4096] * 7      Increase std of initial weights  
hs = []                from 0.01 to 0.05  
x = np.random.randn(16, dims[0])  
for Din, Dout in zip(dims[:-1], dims[1:]):  
    W = 0.05 * np.random.randn(Din, Dout)  
    x = np.tanh(x.dot(W))  
    hs.append(x)
```

All activations saturate

Q: What do the gradients look like?



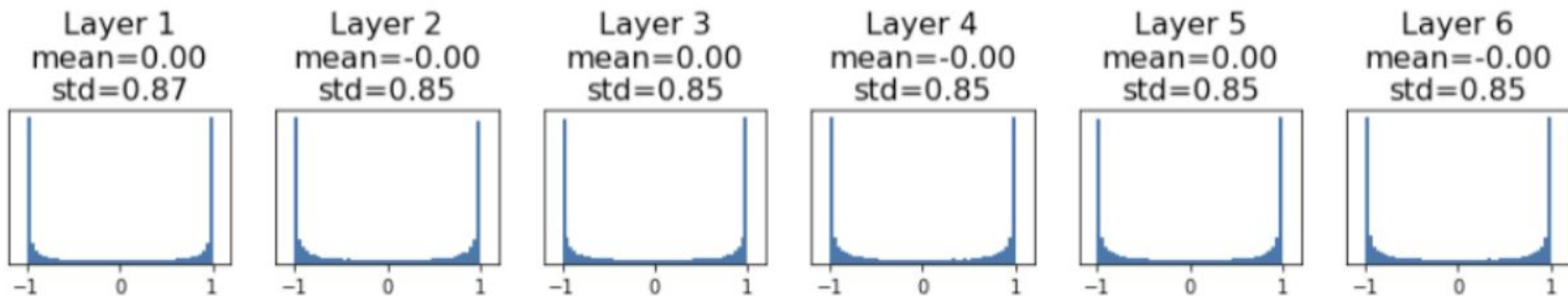
Weight Initialization: Activation Statistics

```
dims = [4096] * 7      Increase std of initial weights  
hs = []                from 0.01 to 0.05  
x = np.random.randn(16, dims[0])  
for Din, Dout in zip(dims[:-1], dims[1:]):  
    W = 0.05 * np.random.randn(Din, Dout)  
    x = np.tanh(x.dot(W))  
    hs.append(x)
```

All activations saturate

Q: What do the gradients look like?

A: Local gradients all zero, no learning =(



Weight Initialization: Xavier Initialization

```
dims = [4096] * 7          "Xavier" initialization:
hs = []                    std = 1/sqrt(Din)
x = np.random.randn(16, dims[0])
for Din, Dout in zip(dims[:-1], dims[1:]):
    W = np.random.randn(Din, Dout) / np.sqrt(Din)
    x = np.tanh(x.dot(W))
    hs.append(x)
```

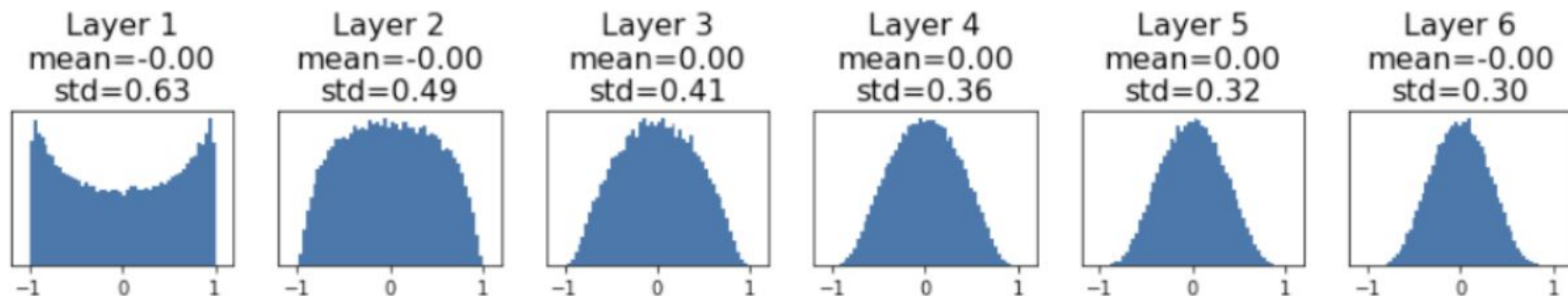
Glorot and Bengio, "Understanding the difficulty of training deep feedforward neural networks", AISTAT 2010

Weight Initialization: Xavier Initialization

```
dims = [4096] * 7          "Xavier" initialization:
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    W = np.random.randn(Din, Dout) / np.sqrt(Din)
    x = np.tanh(x.dot(W))
    hs.append(x)
```

“Just right”: Activations are nicely scaled for all layers!

For conv layers, Din is $\text{kernel_size}^2 * \text{input_channels}$



Glorot and Bengio, “Understanding the difficulty of training deep feedforward neural networks”, AISTAT 2010

Weight Initialization: Xavier Initialization

“Xavier” initialization:
 $\text{std} = 1/\sqrt{\text{Din}}$

Derivation: Variance of output = Variance of input

$$y = Wx$$

$$y_i = \sum_{j=1}^{\text{Din}} x_j w_j$$

$$\text{Var}(y_i) = \text{Din} * \text{Var}(x_i w_i)$$

[Assume x, w are iid]

$$= \text{Din} * (E[x_i^2]E[w_i^2] - E[x_i]^2 E[w_i]^2)$$

[Assume x, w independent]

$$= \text{Din} * \text{Var}(x_i) * \text{Var}(w_i)$$

[Assume x, w are zero-mean]

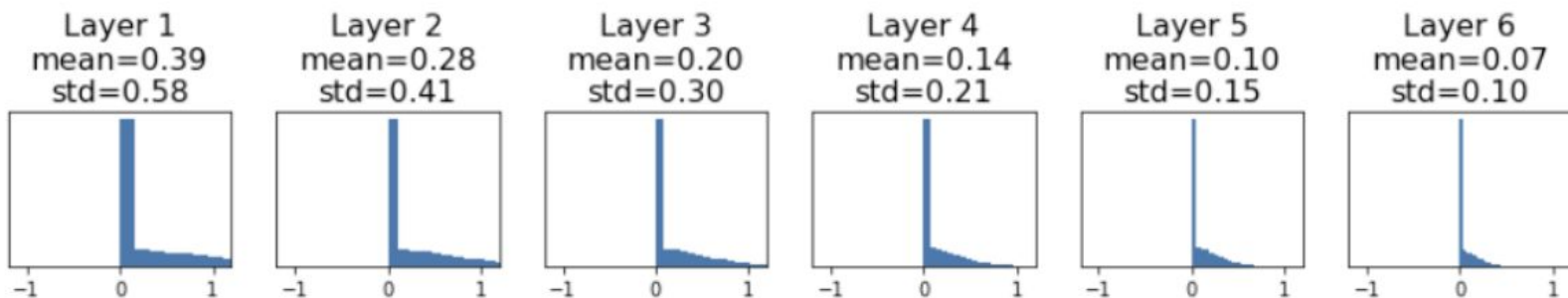
If $\text{Var}(w_i) = 1/\text{Din}$ then $\text{Var}(y_i) = \text{Var}(x_i)$

Weight Initialization: What about ReLU?

```
dims = [4096] * 7      Change from tanh to ReLU
hs = []
x = np.random.randn(16, dims[0])
for Din, Dout in zip(dims[:-1], dims[1:]):
    W = np.random.randn(Din, Dout) / np.sqrt(Din)
    x = np.maximum(0, x.dot(W))
    hs.append(x)
```

Xavier assumes zero centered activation function

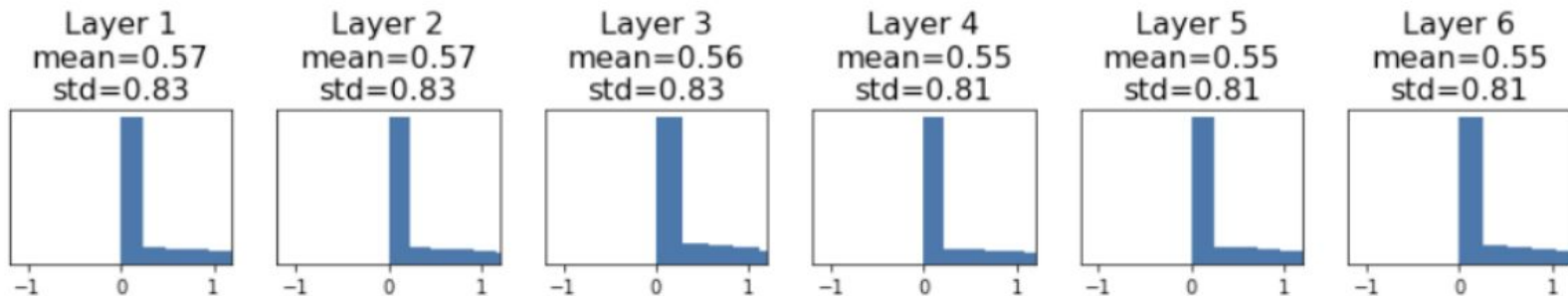
Activations collapse to zero again, no learning =(



Weight Initialization: Kaiming / MSRA Initialization

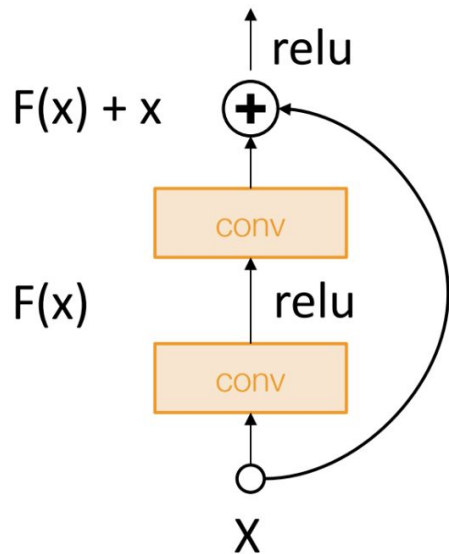
```
dims = [4096] * 7 ReLU correction: std = sqrt(2 / Din)
hs = []
x = np.random.randn(16, dims[0])
for Din, Dout in zip(dims[:-1], dims[1:]):
    W = np.random.randn(Din, Dout) / np.sqrt(Din)
    x = np.maximum(0, x.dot(W))
    hs.append(x)
```

“Just right” – activations nicely scaled for all layers



He et al, “Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification”, ICCV 2015

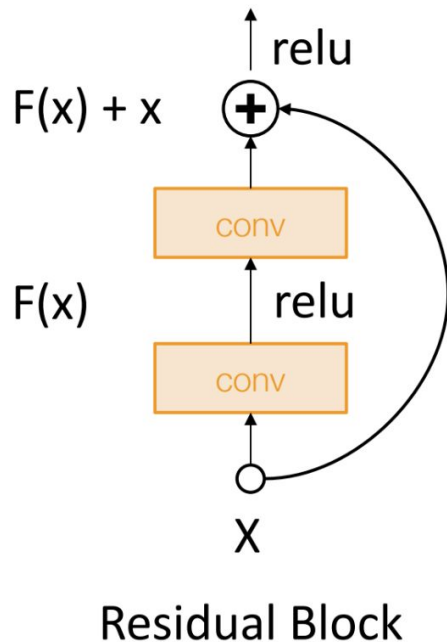
Weight Initialization: Residual Networks



Residual Block

If we initialize with MSRA: then $\text{Var}(F(x)) = \text{Var}(x)$
But then $\text{Var}(F(x) + x) > \text{Var}(x)$ – variance grows with each block!

Weight Initialization: Residual Networks



If we initialize with MSRA: then $\text{Var}(F(x)) = \text{Var}(x)$
But then $\text{Var}(F(x) + x) > \text{Var}(x)$ – variance grows with each block!

Solution: Initialize first conv with MSRA, initialize second conv to zero. Then $\text{Var}(x + F(x)) = \text{Var}(x)$

Proper initialization: active area of research

Understanding the difficulty of training deep feedforward neural networks

by Glorot and Bengio, 2010

Exact solutions to the nonlinear dynamics of learning in deep linear neural networks by Saxe et al, 2013

Random walk initialization for training very deep feedforward networks by Sussillo and Abbott, 2014

Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification by He et al., 2015

Data-dependent Initializations of Convolutional Neural Networks by Krähenbühl et al., 2015

All you need is a good init, Mishkin and Matas, 2015

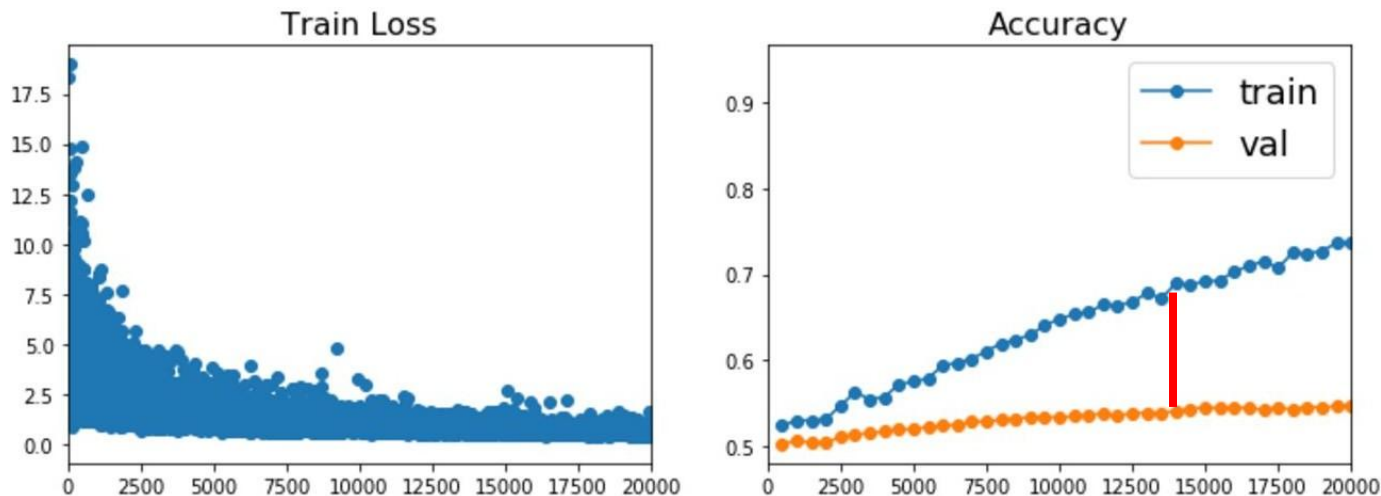
...

Fixup Initialization: Residual Learning Without Normalization, Zhang et al, 2019

The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks, Frankle and Carbin, 2019

Regularization

How to improve single model performance?



Regularization

Regularization: add term to loss

$$L = \frac{1}{N} \sum_{i=1}^N \sum_{j \neq y_i} \max(0, f(x_i; W)_j - f(x_i; W)_{y_i} + 1) + \lambda R(W)$$

In common use:

L2 regularization

$$R(W) = \sum_k \sum_l W_{k,l}^2 \quad (\text{Weight decay})$$

L1 regularization

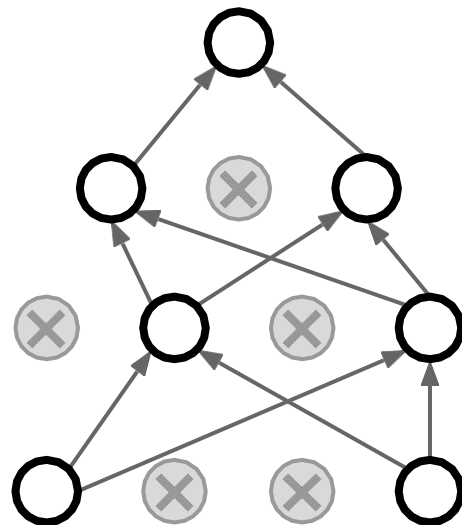
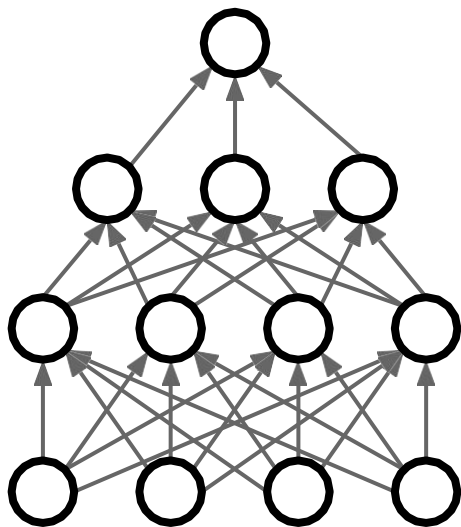
$$R(W) = \sum_k \sum_l |W_{k,l}|$$

Elastic net (L1 + L2)

$$R(W) = \sum_k \sum_l \beta W_{k,l}^2 + |W_{k,l}|$$

Regularization: Dropout

In each forward pass, randomly set some neurons to zero
Probability of dropping is a hyperparameter; 0.5 is common



Srivastava et al, "Dropout: A simple way to prevent neural networks from overfitting", JMLR 2014

Regularization: Dropout

```
p = 0.5 # probability of keeping a unit active. higher = less dropout
```

```
def train_step(X):
```

```
    """ X contains the data """
```

```
    # forward pass for example 3-layer neural network
```

```
    H1 = np.maximum(0, np.dot(W1, X) + b1)
```

```
    U1 = np.random.rand(*H1.shape) < p # first dropout mask
```

```
    H1 *= U1 # drop!
```

```
    H2 = np.maximum(0, np.dot(W2, H1) + b2)
```

```
    U2 = np.random.rand(*H2.shape) < p # second dropout mask
```

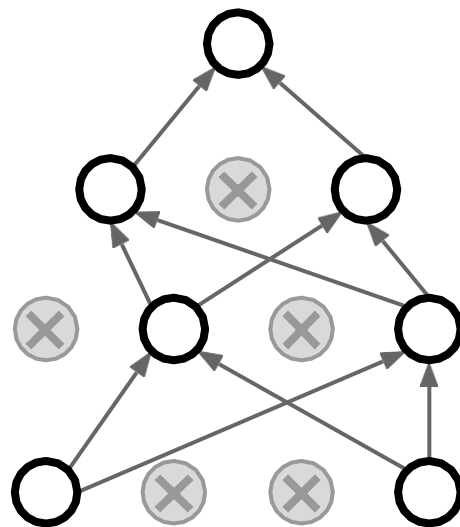
```
    H2 *= U2 # drop!
```

```
    out = np.dot(W3, H2) + b3
```

```
    # backward pass: compute gradients... (not shown)
```

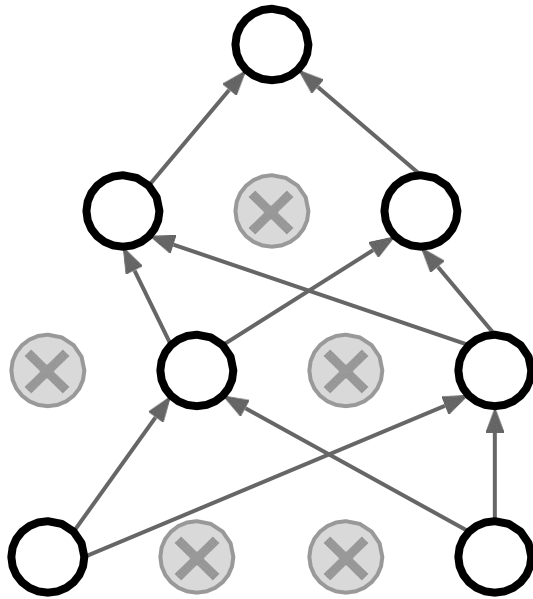
```
    # perform parameter update... (not shown)
```

Example forward pass with a 3-layer network using dropout

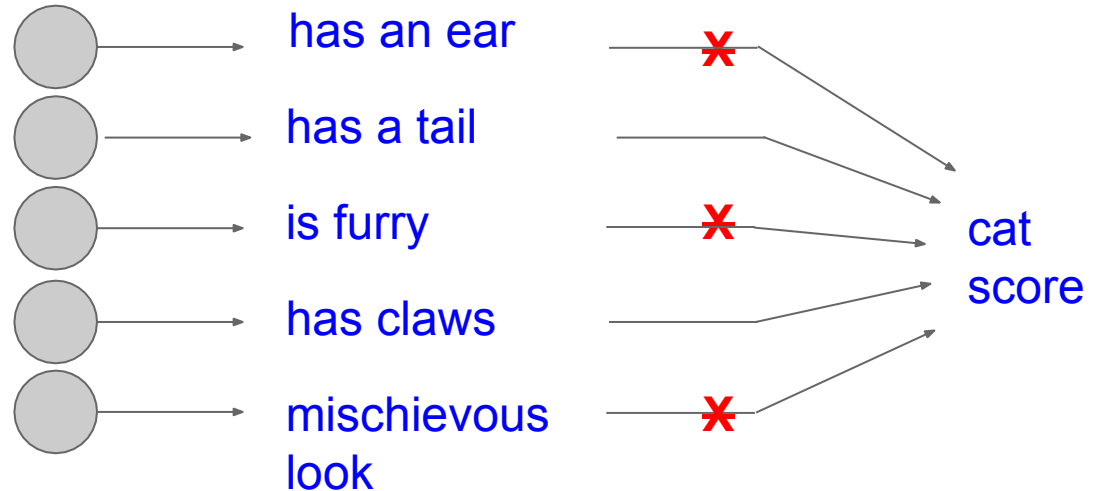


Regularization: Dropout

How can this possibly be a good idea?

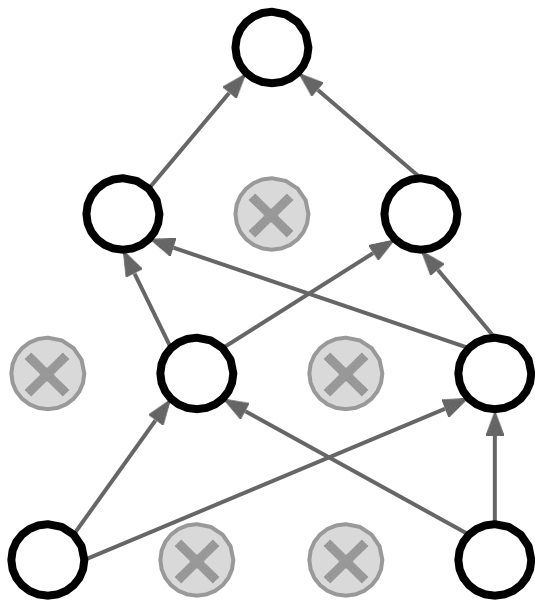


Forces the network to have a redundant representation;
Prevents co-adaptation of features



Regularization: Dropout

How can this possibly be a good idea?



Another interpretation:

Dropout is training a large **ensemble** of models (that share parameters).

Each binary mask is one model

An FC layer with 4096 units has $2^{4096} \sim 10^{1233}$ possible masks!

Only $\sim 10^{82}$ atoms in the universe...

Regularization: Dropout

Dropout makes our output random!

Output (label) Input (image)

$$\boxed{y} = f_W(\boxed{x}, \boxed{z})$$

Random mask

Want to “average out” the randomness at test-time

$$y = f(x) = E_z[f(x, z)] = \int p(z) f(x, z) dz$$

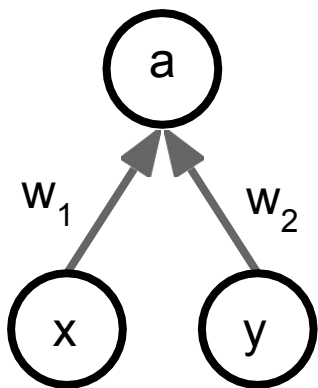
But this integral seems hard ...

Dropout: test time

Want to approximate
the integral

$$y = f(x) = E_z [f(x, z)] = \int p(z) f(x, z) dz$$

Consider a single neuron.



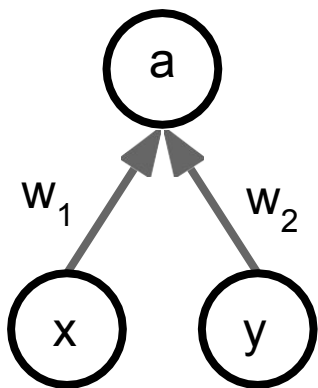
Dropout: test time

Want to approximate
the integral

$$y = f(x) = E_z[f(x, z)] = \int p(z)f(x, z)dz$$

Consider a single neuron.

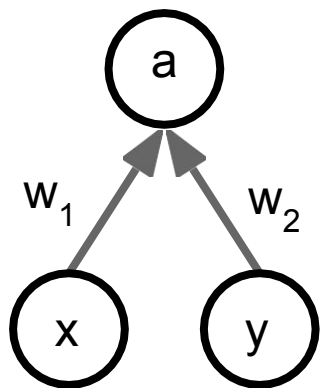
At test time we have: $E[a] = w_1x + w_2y$



Dropout: test time

Want to approximate
the integral

$$y = f(x) = E_z[f(x, z)] = \int p(z)f(x, z)dz$$



Consider a single neuron.

At test time we have: $E[a] = w_1x + w_2y$

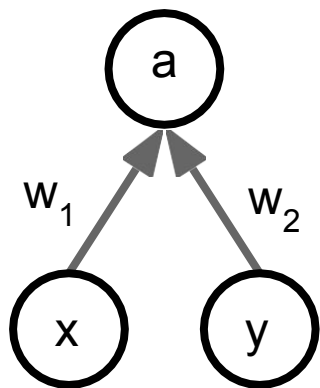
During training we have:

$$\begin{aligned} E[a] &= \frac{1}{4}(w_1x + w_2y) + \frac{1}{4}(w_1x + 0y) \\ &\quad + \frac{1}{4}(0x + 0y) + \frac{1}{4}(0x + w_2y) \\ &= \frac{1}{2}(w_1x + w_2y) \end{aligned}$$

Dropout: test time

Want to approximate
the integral

$$y = f(x) = E_z [f(x, z)] = \int p(z) f(x, z) dz$$



Consider a single neuron.

At test time we have: $E[a] = w_1x + w_2y$

During training we have:

$$\begin{aligned} E[a] &= \frac{1}{4}(w_1x + w_2y) + \frac{1}{4}(w_1x + 0y) \\ &\quad + \frac{1}{4}(0x + 0y) + \frac{1}{4}(0x + w_2y) \\ &= \frac{1}{2}(w_1x + w_2y) \end{aligned}$$

**At test time, multiply
by dropout probability**

Dropout: test time

```
def predict(X):  
    # ensembled forward pass  
    H1 = np.maximum(0, np.dot(W1, X) + b1) * p # NOTE: scale the activations  
    H2 = np.maximum(0, np.dot(W2, H1) + b2) * p # NOTE: scale the activations  
    out = np.dot(W3, H2) + b3
```

At test time all neurons are active always

=> We must scale the activations so that for each neuron:

output at test time = expected output at training time

Dropout Summary

```
""" Vanilla Dropout: Not recommended implementation (see notes below) """
```

```
p = 0.5 # probability of keeping a unit active. higher = less dropout
```

```
def train_step(X):
```

```
    """ X contains the data """
```

```
    # forward pass for example 3-layer neural network
```

```
    H1 = np.maximum(0, np.dot(W1, X) + b1)
```

```
    U1 = np.random.rand(*H1.shape) < p # first dropout mask
```

```
    H1 *= U1 # drop!
```

```
    H2 = np.maximum(0, np.dot(W2, H1) + b2)
```

```
    U2 = np.random.rand(*H2.shape) < p # second dropout mask
```

```
    H2 *= U2 # drop!
```

```
    out = np.dot(W3, H2) + b3
```

```
    # backward pass: compute gradients... (not shown)
```

```
    # perform parameter update... (not shown)
```

```
def predict(X):
```

```
    # ensembled forward pass
```

```
    H1 = np.maximum(0, np.dot(W1, X) + b1) * p # NOTE: scale the activations
```

```
    H2 = np.maximum(0, np.dot(W2, H1) + b2) * p # NOTE: scale the activations
```

```
    out = np.dot(W3, H2) + b3
```

drop in forward pass

scale at test time

More common: “inverted dropout”

```
p = 0.5 # probability of keeping a unit active. higher = less dropout

def train_step(X):
    # forward pass for example 3-layer neural network
    H1 = np.maximum(0, np.dot(W1, X) + b1)
    U1 = (np.random.rand(*H1.shape) < p) / p # first dropout mask. Notice /p!
    H1 *= U1 # drop!
    H2 = np.maximum(0, np.dot(W2, H1) + b2)
    U2 = (np.random.rand(*H2.shape) < p) / p # second dropout mask. Notice /p!
    H2 *= U2 # drop!
    out = np.dot(W3, H2) + b3

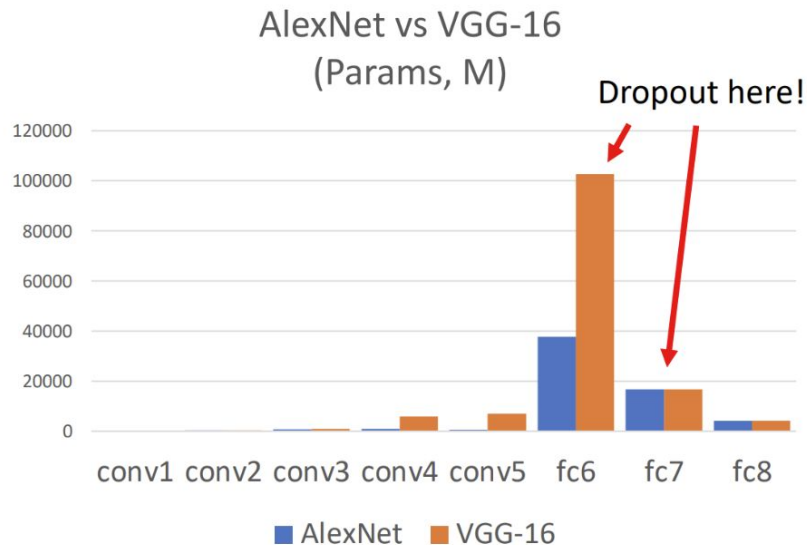
    # backward pass: compute gradients... (not shown)
    # perform parameter update... (not shown)

def predict(X):
    # ensembled forward pass
    H1 = np.maximum(0, np.dot(W1, X) + b1) # no scaling necessary
    H2 = np.maximum(0, np.dot(W2, H1) + b2)
    out = np.dot(W3, H2) + b3
```

test time is unchanged!

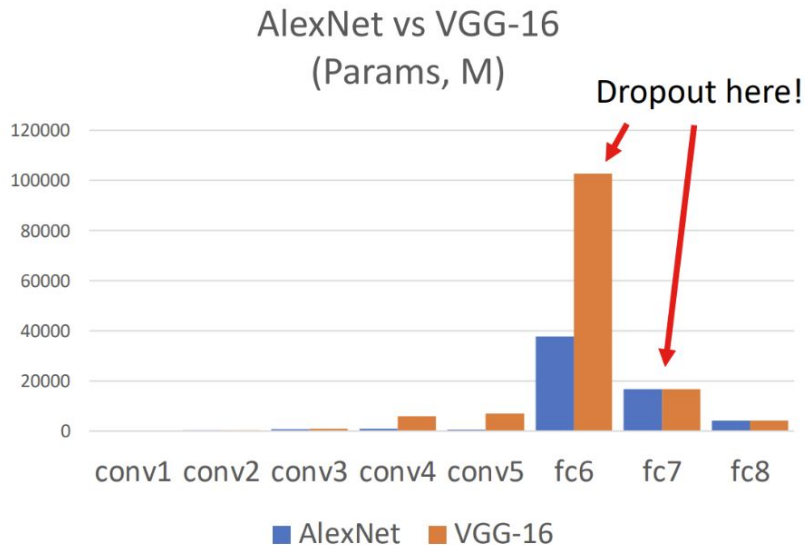
Dropout architectures

Recall AlexNet, VGG have most of their parameters in **fully-connected layers**; usually Dropout is applied there



Dropout architectures

Recall AlexNet, VGG have most of their parameters in **fully-connected layers**; usually Dropout is applied there



Later architectures (GoogLeNet, ResNet, etc) use global average pooling instead of fully-connected layers: they don't use dropout at all!

Regularization: a common pattern

Training: Add some kind of randomness

$$y = f_W(x, z)$$

Testing: Average out randomness (sometimes approximate)

$$y = f(x) = E_z[f(x, z)] = \int p(z)f(x, z)dz$$

Example: Batch Normalization

Training:
Normalize using stats from random minibatches

Testing: Use fixed stats to normalize

Regularization: a common pattern

Training: Add some kind of randomness

$$y = f_W(x, z)$$

For ResNet and later,
often L2 and Batch
Normalization are
the only regularizers!

Testing: Average out randomness
(sometimes approximate)

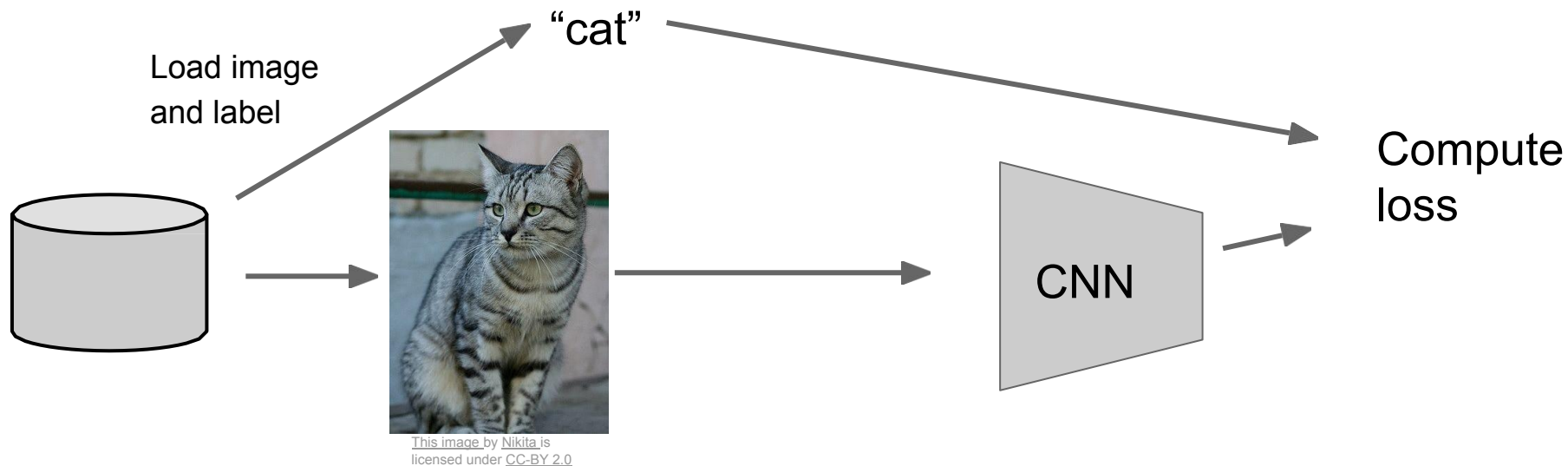
$$y = f(x) = E_z[f(x, z)] = \int p(z)f(x, z)dz$$

Example: Batch
Normalization

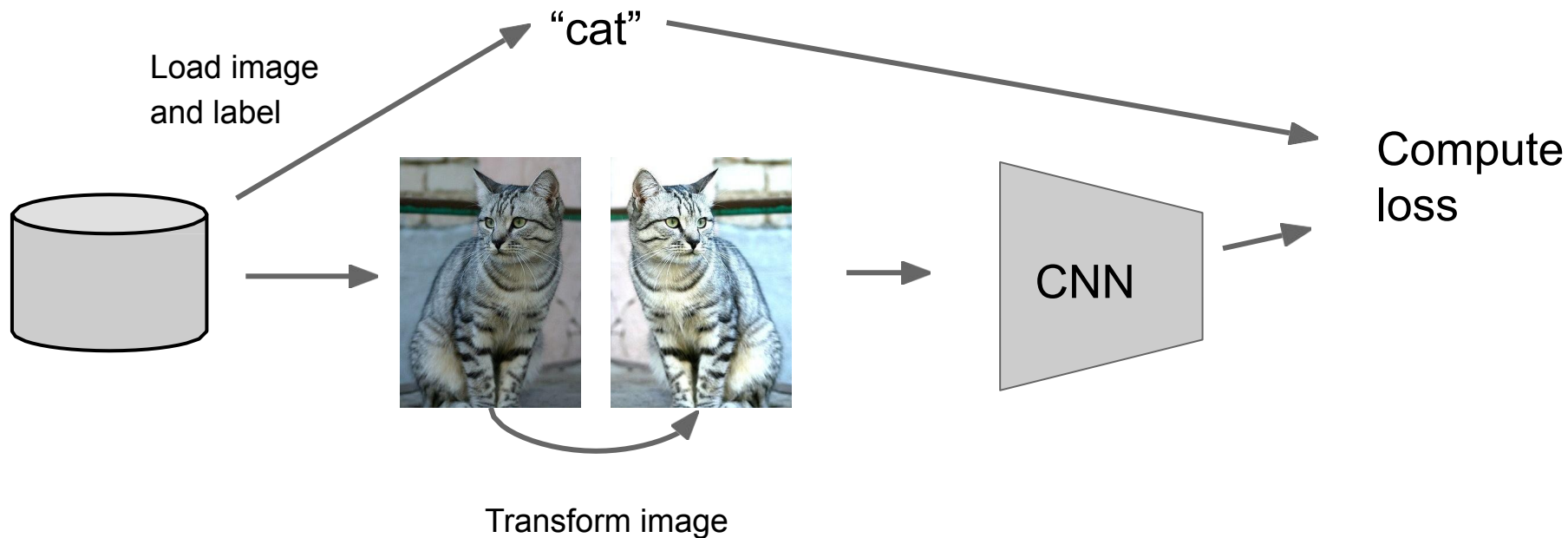
Training:
Normalize using
stats from random
minibatches

Testing: Use fixed
stats to normalize

Regularization: Data Augmentation



Regularization: Data Augmentation



Regularization: Data Augmentation

Horizontal Flips



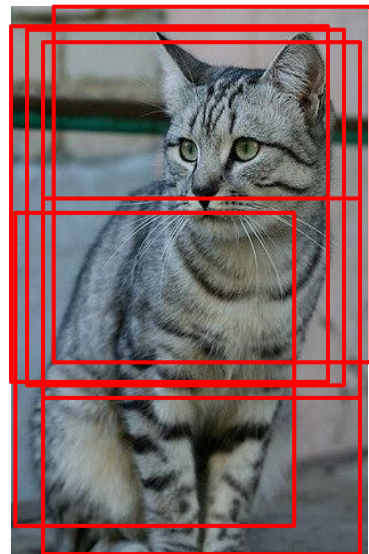
Regularization: Data Augmentation

Random crops and scales

Training: sample random crops / scales

ResNet:

1. Pick random L in range $[256, 480]$
2. Resize training image, short side = L
3. Sample random 224×224 patch



Regularization: Data Augmentation

Random crops and scales

Training: sample random crops / scales

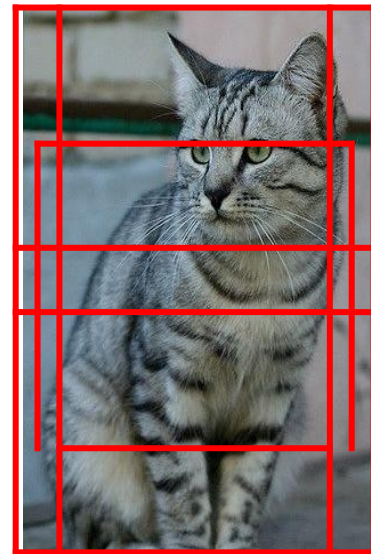
ResNet:

1. Pick random L in range $[256, 480]$
2. Resize training image, short side = L
3. Sample random 224×224 patch

Testing: average a fixed set of crops

ResNet:

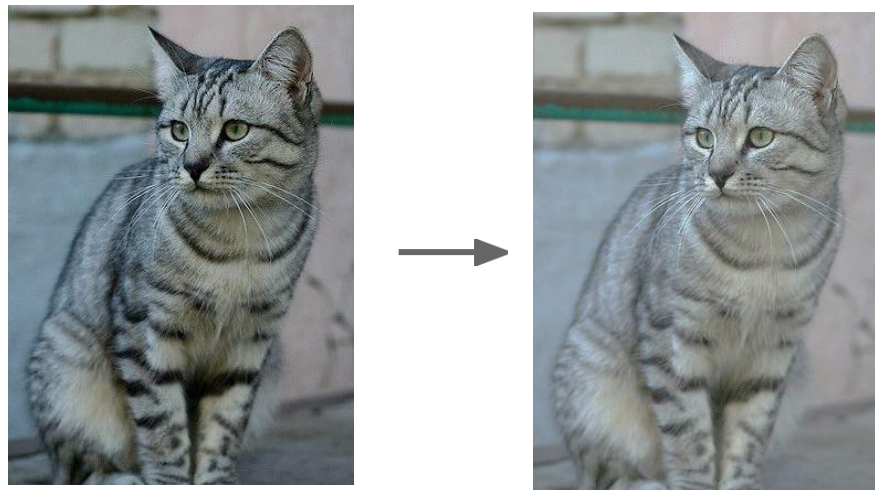
1. Resize image at 5 scales: $\{224, 256, 384, 480, 640\}$
2. For each size, use 10 224×224 crops: 4 corners + center, + flips



Regularization: Data Augmentation

Color Jitter

Simple: Randomize
contrast and brightness



Regularization: Data Augmentation

Color Jitter

Simple: Randomize contrast and brightness



More Complex:

1. Apply PCA to all [R, G, B] pixels in training set
2. Sample a “color offset” along principal component directions
3. Add offset to all pixels of a training image

(As seen in [Krizhevsky et al. 2012], ResNet, etc)

Regularization: Data Augmentation

Get creative for your problem!

Random mix/combinations of :

- translation
- rotation
- stretching
- shearing,
- lens distortions, ... (go crazy)

Regularization: A common pattern

Training: Add random noise

Testing: Marginalize over the noise

Examples:

Dropout

Batch Normalization

Data Augmentation

Regularization: A common pattern

Training: Add random noise

Testing: Marginalize over the noise

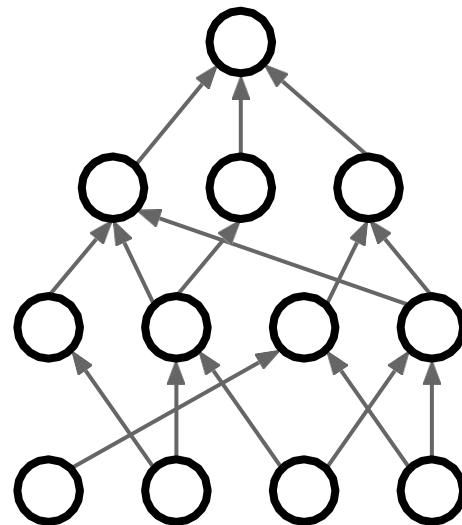
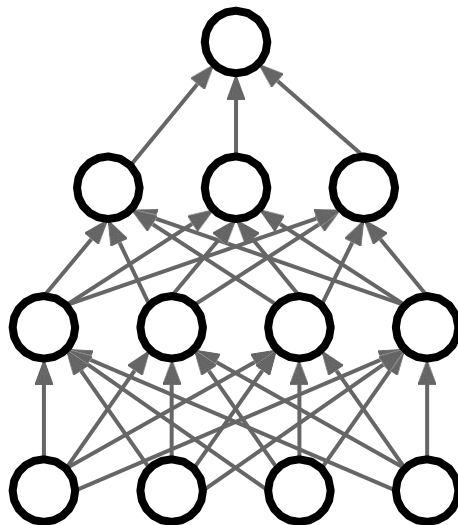
Examples:

Dropout

Batch Normalization

Data Augmentation

DropConnect



Wan et al, "Regularization of Neural Networks using DropConnect", ICML 2013

Regularization: A common pattern

Training: Add random noise

Testing: Marginalize over the noise

Examples:

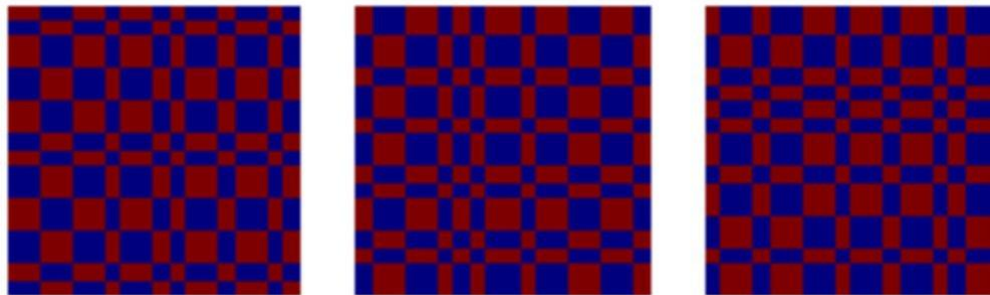
Dropout

Batch Normalization

Data Augmentation

DropConnect

Fractional Max Pooling



Graham, "Fractional Max Pooling", arXiv 2014

Regularization: A common pattern

Training: Add random noise

Testing: Marginalize over the noise

Examples:

Dropout

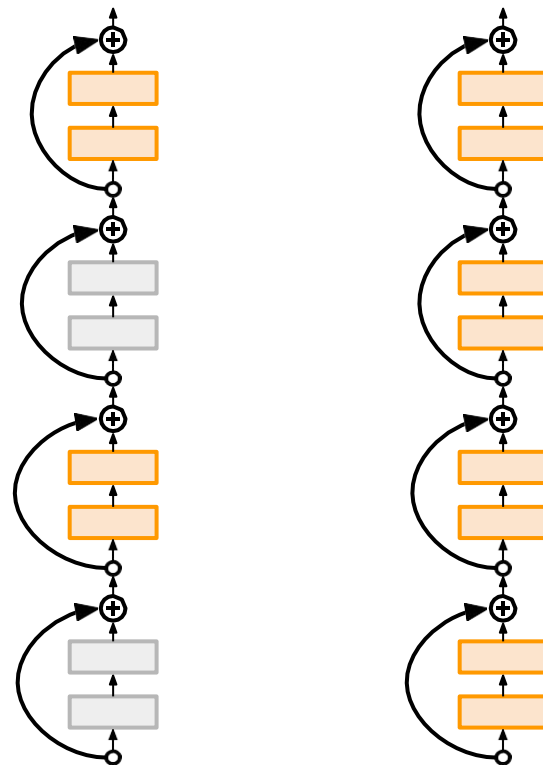
Batch Normalization

Data Augmentation

DropConnect

Fractional Max Pooling

Stochastic Depth



Huang et al, "Deep Networks with Stochastic Depth", ECCV 2016

Regularization: Stochastic Depth

Training: Set random images regions to 0

Testing: Use the whole image

Examples:

Dropout

Batch Normalization

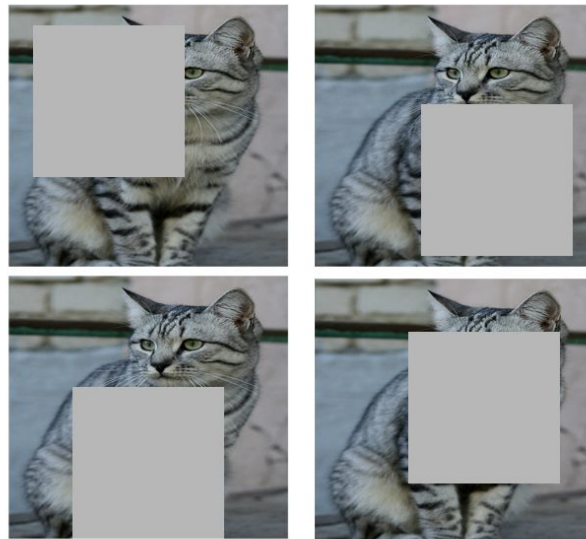
Data Augmentation

DropConnect

Fractional Max Pooling

Stochastic Depth

Cutout



Works very well for small datasets like CIFAR, less common for large datasets like ImageNet

DeVries and Taylor, "Improved Regularization of Convolutional Neural Networks with Cutout", arXiv 2017

Regularization: Mixup

Training: Train on random blends of images

Testing: Use original images

Examples:

Dropout

Batch Normalization

Data Augmentation

DropConnect

Fractional Max Pooling

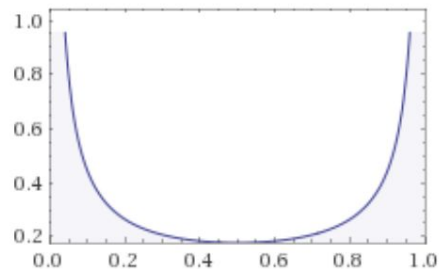
Stochastic Depth

Cutout

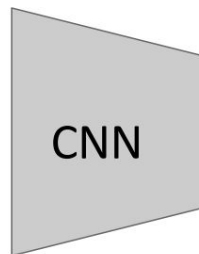
Mixup



Randomly blend the pixels of pairs of training images, e.g. 40% cat, 60% dog



Sample blend probability from a beta distribution $\text{Beta}(a, b)$ with $a=b \approx 0$ so blend weights are close to 0/1



Target label:
cat: 0.4
dog: 0.6

Regularization: Mixup

Training: Train on random blends of images

Testing: Use original images

Examples:

Dropout

Batch Normalization

Data Augmentation

DropConnect

Fractional Max Pooling

Stochastic Depth

Cutout

Mixup

- Consider dropout for large fully-connected layers
- Batch normalization and data augmentation almost always a good idea
- Try cutout and mixup especially for small classification datasets

Training Neural Networks: Overview

1. One time setup

activation functions, preprocessing, weight initialization, regularization

2. Training dynamics

babysitting the learning process, parameter updates, hyperparameter optimization

3. After training

model ensembles, transfer learning

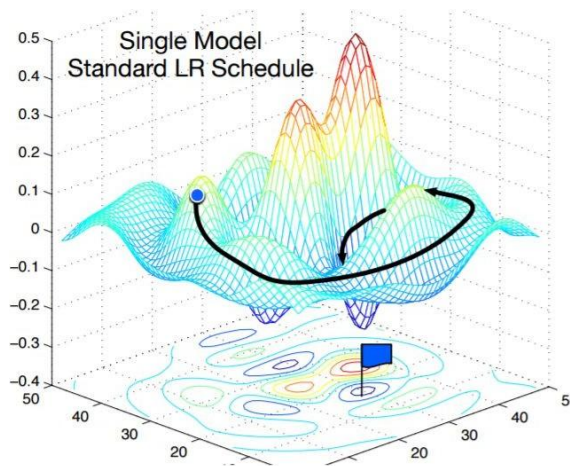
Model Ensembles

1. Train multiple independent models
2. At test time average their results

Enjoy 2% extra performance

Model Ensembles: Tips and Tricks

Instead of training independent models, use multiple snapshots of a single model during training!



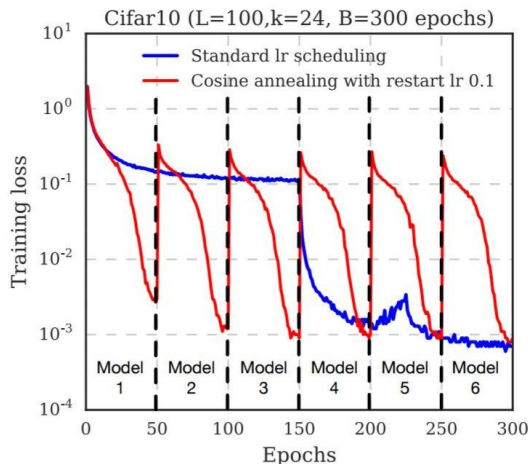
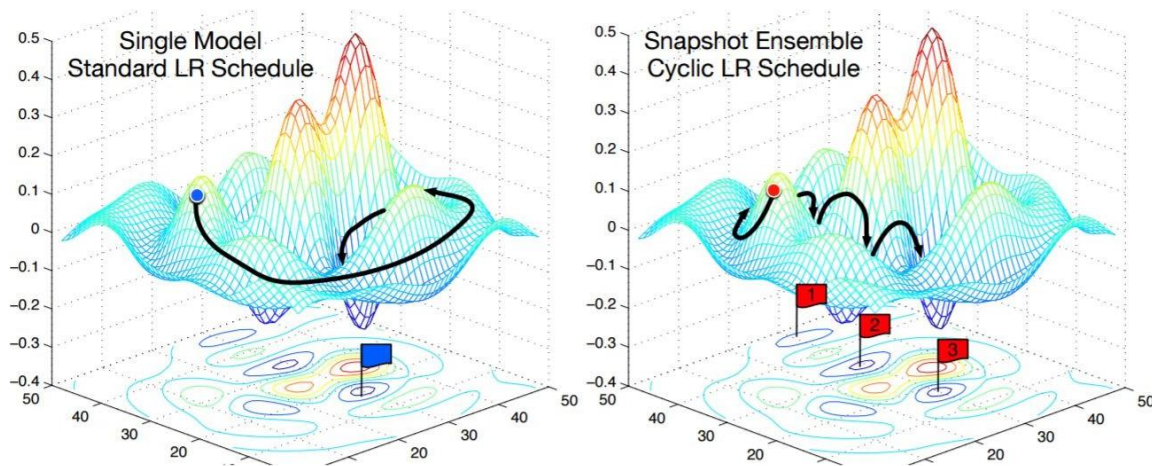
Loshchilov and Hutter, "SGDR: Stochastic gradient descent with restarts", arXiv 2016

Huang et al, "Snapshot ensembles: train 1, get M for free", ICLR 2017

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Model Ensembles: Tips and Tricks

Instead of training independent models, use multiple snapshots of a single model during training!



Cyclic learning rate schedules can make this work even better!

Loshchilov and Hutter, "SGDR: Stochastic gradient descent with restarts", arXiv 2016
Huang et al, "Snapshot ensembles: train 1, get M for free", ICLR 2017
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Model Ensembles: Tips and Tricks

Instead of using actual parameter vector, keep a moving average of the parameter vector and use that at test time (Polyak averaging)

```
while True:
    data_batch = dataset.sample_data_batch()
    loss = network.forward(data_batch)
    dx = network.backward()
    x += - learning_rate * dx
    x_test = 0.995*x_test + 0.005*x # use for test set
```

Polyak and Juditsky, "Acceleration of stochastic approximation by averaging", SIAM Journal on Control and Optimization, 1992.